Learning Traps

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Abstract

When constituents are uncertain about whether leaders' policy preferences align with their interests, policy choices generate a tradeoff between payoffs and information. Leaders can experiment with their preferred policy, but if the outcome is bad, constituents learn this and threaten office removal. I use a repeated model of political agency to analyze how this tradeoff shapes long-lived, term-unlimited leaders' policy choices. When there is uncertainty about the optimal policy for constituents, leaders implement "learning traps": spatially intermediate policies that halt learning about what the best policy is. They experiment only when their preferences are very likely to be aligned with constituents, a contrast to the "gambling for resurrection" logic ubiquitous in the literature. I provide a case study that examines how the 1861 reforms ending serfdom in Imperial Russia represented the logic of a "learning trap" and illustrate additional applications of the model to political contagion and economic policymaking.

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1 Introduction

Leaders and their followers alike are often unsure about what policies serve followers' interests. A mayor may want to expand the police, but her constituents may not know whether they will benefit from more public safety, and thus whether it is worth the extra spending. Or, a hawkish CEO may want to cut the workforce, but the board of directors may not know if it would boost profits. These situations present several considerations: what policies a leader prefers, whether followers can replace her, and what followers learn from her choices.

How are leaders' policy choices shaped by followers' perceptions about the right thing to do? When the effects of policies are *certain*, leaders often pursue their most preferred policy that keeps them in office (Acemoglu and Robinson, 2000, 2001; De Mesquita and Smith, 2010; Boix and Svolik, 2013; Li et al., 2014; Dower et al., 2018). For example, if constituents view police spending as wasteful, our mayor would choose the highest level of spending that keeps them indifferent to voting her out. However, if followers are *uncertain* about their alignment with the leader's preferences, policies generate *information*. If the mayor went all-in with funding the police and citizens experienced its benefits, they would allow her to sustain spending. If it backfired, they would constrain spending or even replace her. *Partially* funding the project might be safer: inefficiencies could be consistent with policing being valuable and underfunded or wasteful and overfunded. Even if constituents hold (heterogeneous) opinions about a policy's effects, their estimates may carry uncertainty, and what they learn depends on what the leader implements.

Many political agency models examine how these tradeoffs compel term-limited, "short-lived" leaders to *gamble for resurrection*. When popular, they pursue safe policies to ensure they can implement their preferred policy before leaving office, and gamble with risky policies when at risk of replacement. This logic has been applied to reform (Dur, 2001; Fu and Li, 2014; Dewan and Hortala-Vallve, 2019), conflict (Downs and Rocke, 1994), and voter learning (Tomasi, 2023; Izzo, 2024; Bils and Izzo, 2025). Yet, it is less applicable to "long-lived" leaders like dictators or entrenched elected officials, who often lack term limits and hence continually face threats of replacement. Moreover, leaders who have consolidated power may avoid replacement by conceding to followers. Thus, how are long-lived leaders' policies affected by followers' perceptions differently from short-lived ones?

This paper answers these questions with a repeated model of political agency. It shows that when there is uncertainty about which policies benefit constituents, long-lived leaders pursue moderate, middle-ground policies called *learning traps* that prevent constituents from learning what policies serve their interests. Contrary to the gambling for resurrection logic, long-lived leaders experiment when they are viewed favorably. When viewed unfavorably, they concede to constituents' demands instead of "gambling." They halt learning only when constituents are uncertain about whether their preferences are aligned, and are thus viewed relatively neutrally.

Because long-lived leaders' choices affect their actions at every point in the future, they trade off between perpetually implementing learning traps — guaranteeing a moderate payoff — or revealing information. When constituents view a leader's preferred policy favorably, information is likely to favor her, so she experiments with that policy. When its effects are uncertain, experimentation could backfire, creating an incentive to shut down information. If the policy is likely to harm constituents, they force her to pursue policies closer to their interests. Leaders thus halt learning when the effects of policies are uncertain and information would be most valuable.

Can we use the model to study phenomena like reform and conflict? Using a case study, I first illustrate how the model's insights can explain why 1860s Russia did not adopt liberal economic institutions like Western Europe. I argue that while the Russian government preferred these institutions, they were uncertain about whether they would be aligned with everyday Russians' interests — in part because Russia was culturally different from Western Europe — and consequently pursued an inefficient middle ground that mixed elements of liberal and communal institutions. I show how the internal logic shaping the government's policy choices is consistent with the notion of a learning trap. Next, I extend the model to study policy contagion and political fragmentation by comparing 19th-century Europe's experiments with indirect rule to China's lack thereof. I finish with additional examples of learning traps, including the Soviet refusal to adopt "full communism," FDR's unwillingness to include healthcare reform in the New Deal, and the US's involvement in "limited wars."

Theoretical Overview In the model, policies are denoted x, and the leader prefers a higher policy. A representative for "the people" wants to maximize an outcome y, and can overthrow the leader. There are two states of the world: a "pro-leader" state where the relationship between

x and y is positive — meaning the leader's preferred policy is optimal for the people — and an "anti-leader" state where it is negative, meaning the leader's preferences are misaligned with the people. The leader and people share a common prior belief. As different policies are implemented, information is revealed. The leader and people use histories of policy-outcome data (y, x) to infer the state. The key feature is that these data are generated endogenously: by setting x, the leader also controls inference. Information revolves around a "learning trap" policy, which is *spatially intermediate* and, when implemented, generates no information about what the optimal policy is.[1]

The intuition for the model's main insight — leaders implement learning traps when beliefs in the "pro-leader" state are intermediate — is as follows. In a repeated setting, a leader chooses between implementing the learning trap for all time — perpetually shutting down information but receiving a modest payoff — or experimenting with her preferred policy. Experimentation could work in her favor — in which case she can forever pursue her preferred policy — but could backfire — in which case the people will force a low policy. When the people believe the pro-leader state is more likely, revealing information favors the leader in expectation, creating an incentive to experiment with a high policy. When the people are unsure about the state, the leader implements the learning trap, since experimentation could tie her hands if the anti-leader state materializes. When the people believe the anti-leader state is more likely, they threaten overthrow and force the leader to pursue a low policy despite her preference for shutting down learning.

This difference with the "gambling for resurrection" literature stems from two assumptions. First, the present paper's leader is *infinitely-lived*. Term-limited leaders always pursue their preferred policies in their final term. At high beliefs, they thus halt learning to preserve this option for the end of their tenure. In repeated settings, leaders choose between a moderate payoff that reveals no information or an extreme policy that reveals information. Information revelation is only preferred at high beliefs, where it delivers more than a moderate payoff in expectation.

Second, leaders in "gambling for resurrection" models can be replaced by outside options. Leaders only secure retention at low beliefs if they experiment and get lucky. In the present paper, constituents' ability to overthrow leaders depends on their information, i.e. their understanding of alternatives to the status quo. The model is thus better-suited to settings where uncertainty concerns the effects of policies, as opposed to idiosyncratic features of a leader, like her ability.

[1] I show that such a policy not only exists but is spatially intermediate under a simple technical condition.

I show that the leader's attraction to learning traps at intermediate beliefs holds even if the people have a preference for learning the state of the world, and that this may generate further nonmonotonicities in a leader's behavior. This connects the paper to a theoretical literature on the reputational motives of patient agents, such as Holmström (1999), where managers oversupply effort to bias firms' beliefs about their ability. Non-learning results in these models often depend on asymmetries in costs (Manso, 2011), information (Ely and Välimäki, 2003), or signal interpretation (Aghion and Jackson, 2016).[2] The present paper finds that more patient leaders prefer to slow learning since *future* policy constraints become more salient, the opposite of Besley and Case (1995) and Banks and Sundaram (1998). I also discuss how the model's findings generalize to alternative information structures and *ambiguity* about constituent beliefs.

This paper adds to a literature studying how endogenous policy-outcome data may distort voter learning (Levy, 2014; Spiegler, 2016; Eliaz and Spiegler, 2020; Levy and Razin, 2021; Schwartzstein and Sunderam, 2021; Montiel Olea et al., 2022; Hwang, 2023). Voters learning from spatial platforms may converge on inefficient policies (Callander, 2011). Frequentist inference or bounded memory can cause polarization (Levy et al., 2022; Izzo et al., 2023; Levy and Razin, 2025). Less information about policies can be better for voters (Prat, 2005) or politicians (Kartik et al., 2015). Substitutability between politician effort and ability (Ashworth et al., 2017) can hamper learning about ability. The present paper's insights are related to partisan traps (de Mesquita and Dziuda, 2024), where voters elect partisans who obfuscate the gains from common-value policies.

Applications to Land Reform in Russia and Political Contagion To illustrate how the mechanisms of the model can influence the policy choices of autocratic regimes, I study how learning traps played a role in the 1861 reforms ending Serfdom in Russia. Before 1861, Russian peasants (serfs) worked noble land as part of communes. After Russia's 1856 defeat in the Crimean War, many in the Tsar's government believed liberal reforms — like private property and freedom of labor movement — could stimulate economic growth, reduce the power of the nobility (Moon, 2014), jumpstart an industrial revolution (Pereira, 1980), and increase fiscal revenue (Dennison, 2020, 2023). However, they worried liberal policies could backfire, leading to exploitation, fiscal

[2] In the latter, constituents threaten to replace incumbents until a high-ability leader emerges, forcing learning in equilibrium. However, this mechanism is driven by the "outside option" phenomenon, which cannot be replicated in the present paper.

straits, and overthrow (Dennison, 2014; Polunov et al., 2015). That is, there was uncertainty about whether the government's preferred policy benefited constituents.

In the ensuing reforms, the government pursued a middle ground: it mixed private and communal decision-making, distorting investment and production decisions (Dennison, 2020); restricted labor mobility; and inhibited urban migration and industrial development (Nafziger, 2010). I argue that this arrangement was consistent with a learning trap: the government feared unrest if liberal policies backfired but, by pursuing an intermediate policy, distorted economic outcomes to halt learning. The Supporting Information Appendix discusses how this explanation adds to the predominant explanations of the 1861 reforms: state capacity and elite bargaining. The application also connects the model to gradualism in "divide-the-dollar" settings like land reform, as discussed in Roland (2002) and Acemoglu and Robinson (2008), showing that under uncertainty about the optimal form of tenure, mixing different property regimes can obfuscate which is better.

Finally, I extend the model with *exogenous external information revelation* so that leaders' decisions are not the only sources of information about the effects of policies. I show that leaders are less likely to implement learning traps; they cannot totally halt learning and hope experimentation with their preferred policy will favor them. This allows me to study policy "contagion," where countries with similar institutions learn from each other's experiences, showing that it explains why Europe experimented with more indirect rule than China in the 19th century. This connects the paper to empirical work that documents inferences about policy efficacy, including trust in government (Chen and Yang, 2019), external validity of policy experiments (Wang and Yang, 2021), and policy contagion (Buera et al., 2011; Mukand and Rodrik, 2005). I finish by discussing applications of learning traps to the USSR, FDR's refusal to reform healthcare, and why countries like the US may engage in "limited wars."

Plan of Paper I introduce the model in section two and solve it in section three, where I also discuss theoretical extensions. Section four uses the end of Serfdom in Russia to illustrate how a patient regime may pursue a learning trap. Section five solves the model with exogenous information revelation to study political fragmentation in Europe and China. Section six addresses additional applications, and I conclude in section seven.

2 Model

Policies Consider an infinite-horizon model beginning at $t = 1$. Each period, a policy $x_t \in X = [0, 1]$ is implemented that determines an outcome $y_t \in \mathbb{R}$ realized later that period. For example, y can represent profits, income, or the value of a public good.

Players The first player in the model is the "leader" — an entrenched incumbent, autocrat, or CEO. The second player — "the people" — constrains the leader's power and can be interpreted as a representative for an electorate, board of directors, or elite. Both players are infinitely lived.

The leader's discount factor is δ. Her flow utility is $u_\ell(x_t)$ for $u_\ell(\cdot)$ concave and strictly increasing. Preference for higher x_t could reflect an ideological bent, reputational concerns, or the size of rents she can extract in a setting like land reform.

The people's flow utility is y_t. Their discount factor is 0. This simplifies the model to illuminate its key tradeoffs, but we relax this assumption in Section 3.2.

Actions At $t = 1$, the leader holds power. At the beginning of each period in power, she announces a policy. The people decide whether to incur a one-time cost $c > 0$ to overthrow her. If they overthrow, the people set x_t from t onwards. Overthrow succeeds with probability 1; the leader receives a loss $-L < 0$ and no further flow utility. If they do not overthrow, her announced policy x_t is implemented that period and t moves to $t + 1$.

States Players possess uncertainty over two states of the world governing the relationship between x and y. In the "pro-leader" state, increases in x cause increases in y. In the "anti-leader" state, increases in x cause decreases in y:

$$y_t = -(x_t - 1)^2 + \epsilon_t \equiv f_g(x_t) + \epsilon_t \qquad \text{Pro-Leader} \tag{1}$$

$$y_t = -x_t^2 + \epsilon_t \equiv f_b(x_t) + \epsilon_t \qquad \text{Anti-Leader} \tag{2}$$

$\epsilon_t \sim \mathcal{U}[-\sigma, \sigma]$ is i.i.d. every period, making inference potentially imperfect.[3] f_g and f_b are graphed in Figure 1.

[3] Proposition 4 shows the results are robust to single-peaked distributions of the noise terms.

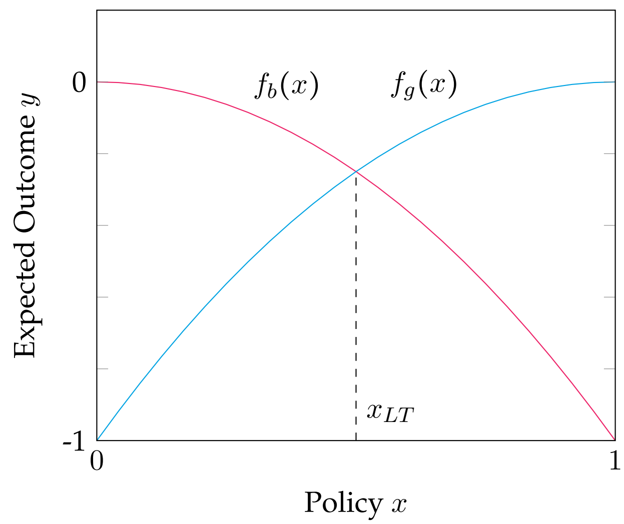

Figure 1: $f_g(x)$ and $f_b(x)$ graphed

The expected value of y conditional on $x = 1$ is higher in the pro-leader than the anti-leader state, and $x = 0$ is better in the anti-leader than the pro-leader state. This payoff symmetry means that $f_g(x) = f_b(x)$ at a policy $x_{LT} = \frac{1}{2}$.

Inference Both players follow Bayes' Rule and use histories $\{y_\tau, x_\tau\}_{\tau \leq t}$ to update a common belief over the two states. Denote q_t the belief in the pro-leader state at the beginning of time t. During period t, players observe a policy-outcome pair (y_t, x_t) and use it to update q_t to q_{t+1}, to be carried into $t+1$. Denote $\Delta(x) = |f_g(x) - f_b(x)|$ the *difference* between the expectation of y conditional on x in each state.

For ease of mathematical exposition, we assume $\sigma \geq \frac{1}{2}$. Because $|f_g(x) - f_b(x)| \leq 1$, $\frac{\Delta(x)}{2\sigma}$ is bounded above by 1. Bayes' Rule implies that the distribution of posteriors conditional on x_t and q_t is:

$$q_{t+1}|q_t, x_t = \begin{cases} 1 & \text{with prob. } \frac{\Delta(x_t)}{2\sigma} q_t \\ q_t & \text{with prob. } 1 - \frac{\Delta(x_t)}{2\sigma} \\ 0 & \text{with prob. } \frac{\Delta(x_t)}{2\sigma}(1 - q_t). \end{cases}$$

When $x_t = x_{LT}$, $f_g = f_b$, meaning $\Delta(x_{LT}) = 0$. x_{LT} shuts down information revelation ($q_t = q_{t+1}$

with probability 1) and we refer to it as a **learning trap policy**. [4] While extreme policies reveal more information, x_{LT} mixes elements of these extremes to muddy inference.

We make two more assumptions. First, we normalize $u_\ell(1) = 1$ and $u_\ell(x_{LT}) = x_{LT}$. This is without loss since utility functions are unique up to positive affine transformations. For example, if utility is linear, $u_\ell(0) = 0$. If utility is strictly concave, $u_\ell(0) < 0$. Second, we assume $c < x_{LT}^2$. This upper bound on the cost of overthrow ensures that if the anti-leader state realizes, the leader cannot implement moderate policies like x_{LT} without risking overthrow.

Sequential Game We describe the game recursively.

1. At time t, if in power, the leader announces a policy x_t. Then, the people decide whether to overthrow the leader.
   (a) If the people decide to overthrow the leader:
      i. The people incur the cost of overthrow c. The leader loses power, receives $-L$, and receives no further flow utility.
      ii. The people set x_t, y_t realizes, and they receive utility y_t.
      iii. The people update q_t to q_{t+1}. Return to step 1(a)ii at $t + 1$.
   (b) If the people decide not to overthrow the leader:
      i. The leader's policy x_t is implemented, y_t realizes, and the people receive utility y_t. The leader in power receives $u_\ell(x_t)$.
      ii. Players update their belief q_t to q_{t+1}. Return to step 1 at $t + 1$.

We focus on pure-strategy Markov Perfect Equilibria with respect to the belief q_t. $x_\ell(q) \in [0, 1]$ indicates the leader's policy plan. The people's strategy is given by $\{o(q), x_p(q)\}$. $o(q) \in \{\text{overthrow}, \text{nooverthrow}\}$ denotes the overthrow decision. $x_p(q) \in [0, 1]$ indicates their policy plan once in power. $x_\ell(q)$ and $\{p(q), x_p(q)\}$ are best responses to each other in equilibrium.

[4] The existence of x_{LT} only depends on the single-crossing property of f_g and f_b, and generalizes beyond uniform noise. If $\epsilon_t \sim p(\epsilon_t)$ for some distribution p :

$$q_{t+1} = \frac{p(y_t - f_g(x_{LT}))q_t}{p(y_t - f_g(x_{LT}))q_t + p(y_t - f_b(x_{LT}))(1 - q_t)} = \frac{p(y_t - f_g(x_{LT}))q_t}{p(y_t - f_g(x_{LT}))q_t + p(y_t - f_g(x_{LT}))(1 - q_t)}$$
$$= \frac{p(y_t - f_g(x_{LT}))q_t}{p(y_t - f_g(x_{LT}))} = q_t$$

Comments The model lends itself to two interpretations of information. The first is policy uncertainty: neither the leader nor people know the effects of policies until they are implemented. The second is that the people utilize one of two subjective models of the world and, correspondingly, subjective *beliefs* to evaluate outcomes: a pro-leader data-generating process ($y_t = f_g(x_t) + \epsilon_t$) and anti-leader process ($y_t = f_b(x_t) + \epsilon_t$). Section 3.3 shows that the model's predictions are similar under *ambiguity* about constituents' beliefs.

The assumption that the people's maximal payoff is identical across states means that learning traps are *spatially intermediate*. The key condition for an intermediate policy to halt learning is thus that extreme policies generate information. This is natural in settings where an incumbent must make a choice about an existing institution or policy. Consider a president who *inherits* a conflict. Continuing or ending it reveals information about her nation's military competency, but continuing it in a limited fashion may reveal less. Since the president inherited the conflict, she does not enjoy the luxury of never starting it (which would traditionally reveal the least information). Similarly, consider a ruler reforming an economic institution, such as land tenure. Because tenure is fundamental to property rights, opting out is impossible. The status quo need not be uninformative in the event of an exogenous institutional or geopolitical shock, and in particular when an ideological shock introduces a novel subjective model mapping policies to outcomes.

What will matter for the results is that the *leader's* equilibrium payoff is higher in the pro-leader than the anti-leader state, and her utility from the *learning trap* is in between. [5] Even if learning trap policies were *extreme* (e.g. $x = 0$), endowing the leader with a reputational loss for pursuing a failed policy could deliver similar results. Payoffs could also take a "regression format" of $y_t = \beta_0 + \beta_1 x_t + \epsilon_t$, where the people believe (β_0, β_1) are somewhere in $\mathbb{R}^2$. Here, the logic of a learning trap could be replicated through *policy variation*. As in Levy and Razin (2025), the people learn the state more quickly if there is a great deal of policy variation, meaning a leader who repeatedly implements the same policy could minimize information.

[5] The insights and most proofs are nearly identical if f_g and f_b are *linear* in x with positive/negative slope, in which case $\Delta(x)$ is still propotional to $|x - x_{LT}|$. Although the people would prefer extreme policies, for intermediate q, the expected utility gap between the learning trap and optimal policy is small.

3 Analysis

We analyze the baseline model in Section 3.1, highlighting how the leader pursues a learning trap at intermediate beliefs in the pro-leader state. Section 3.2 shows how the result generalizes to a setting where the people value both current and future payoffs. Section 3.3 discusses extensions to alternative information structures, non-uniform noise, three states, and ambiguous beliefs.

3.1 Analysis of Baseline Model

People The people's optimal policy, conditional on being in power, is $x_p(q) = \arg\max_x \mathbb{E}[y|x]$; expectations are taken with respect to q. They have a strict incentive to overthrow if and only if

$$\max_x \mathbb{E}[y|x] - c > \mathbb{E}[y|x_\ell(q)] \tag{3}$$

i.e. if the best policy they can achieve is better than the leader's, less the cost of overthrow. In equilibrium, $o(q) = \text{overthrow}$ if and only if (9) is satisfied strictly. The people's optimal policy maximizes $-q(1-x)^2 - (1-q)x^2$, which occurs when $x_p(q) = q$.

For each q, we define $\text{NR}(q)$ as the set of policies that weakly prevents overthrow: $\text{NR}(q) = \{\hat{x} \in [0,1] : \max_x \mathbb{E}[y|x] - c \leq \mathbb{E}[y|\hat{x}]\}$. Because x_{LT} delivers the same utility in both states of the world, there exists a range of interior beliefs where x_{LT} is contained in the interior of $\text{NR}(q)$. Its graph as a function of q is shaded below. $x_p(q)$ is bolded.

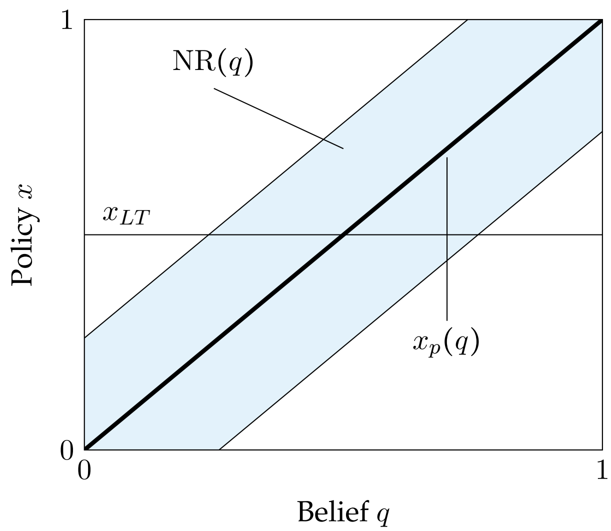

Figure 2: No-overthrow constraint $\mathrm{NR}(q)$

Leader's Solution Let $\overline{x}(q) \equiv \max\{\mathrm{NR}(q)\}$ be the most extreme policy avoiding overthrow. At $q = 0$ or 1, policies have no information externalities. If $q = 1$, the leader implements $x_\ell(1) = \overline{x}(1) = 1$ for all time, and her payoff is $u_\ell(1) = 1$. At $q = 0$, $x_\ell(0) = \overline{x}(0) > 0$ for all time. Because $c < x_{LT}^2$, $\overline{x}(0) < x_{LT}$. We denote $u_\ell(\overline{x}(0)) = \underline{u}$. The key for the result is $1 > x_{LT} > \underline{u}$: the leader's utility at x_{LT} is between that at $q = 1$ and $q = 0$.

Since L is large, the leader will avoid overthrow on the equilibrium path. For interior q, the leader's optimal policy $x_\ell(q)$ solves:

$$V(q) = \max_{x \in \mathrm{NR}(q)} \underbrace{(1-\delta)u_\ell(x)}_{\text{Flow utility}} + \delta\left(\underbrace{\frac{\Delta(x)}{2\sigma}\Psi(q)}_{\text{Truth revealed}} + \underbrace{\left(1 - \frac{\Delta(x)}{2\sigma}\right)V(q)}_{\text{Truth not revealed}} \right) \tag{4}$$

First, we normalize flow utility by $1-\delta$. Next, $\Psi(q) = q + (1-q)\underline{u}$ is the *expected value of revealing the truth*. Implementing a policy x generates information in addition to flow utility. With probability $q\frac{\Delta(x)}{2\sigma}$, $q \to 1$. The leader forever receives flow utility $(1-\delta)\cdot 1$, yielding a total value of 1. With probability $(1-q)\frac{\Delta(x)}{2\sigma}$, $q \to 0$, and the leader's value is $\underline{u}$. Otherwise, q is invariant.

The following proposition characterizes $x_\ell(q)$ for patient leaders.[6]

Proposition 1. *Suppose $\sigma(1-\delta)$ is small or $\underline{u}$ is sufficiently low. There exists a threshold $\underline{q} \in (0,1)$ such*

[6]A more rigorous proof of this proposition is presented in the Supporting Information Appendix: Lemma 8 and Theorem 1.

that:

1. *If* $q \le \underline{q}$, $x_\ell(q) = \min\{x_{LT}, \overline{x}(q)\}$. *The leader plays the learning trap or the closest policy to the learning trap.*

2. *If* $q > \underline{q}$, $x_\ell(q) = \overline{x}(q)$. *The leader plays the maximal policy preventing overthrow.*

Proof. For $q \in (0,1)$, suppose the leader can implement any policy in $[\underline{x}(0), 1]$. We can rearrange the leader's Bellman equation as:

$$\max_{x \in [\underline{x}(0),1]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta)+\delta\Delta(x)} u_\ell(x) + \frac{\delta\Delta(x)}{2\sigma(1-\delta)+\delta\Delta(x)} \Psi(q) \tag{5}$$

This expression is a convex combination of flow utility $u_\ell(x)$ and the value $\Psi(q)$ of the truth. If the leader pursues a strategy where she implements x_{LT} for all time, it delivers a value $u_\ell(x_{LT}) = x_{LT}$. As x moves away from x_{LT}, weight shifts from flow utility $u_\ell(x)$ onto the expected value of revealing information $\Psi(q)$.

Suppose q is low, so that $\Psi(q) < x_{LT}$. If the leader implements $x > x_{LT}$, this results in higher flow utility relative to x_{LT}. But it also reveals information, whose expected value $\Psi(q)$ is worse than x_{LT}. If $\sigma(1-\delta)$ is small, the effect of flow utility is small, meaning that pursuing the learning trap for all time and receiving a modest payoff is better than gaining a bit of flow utility that reveals unfavorable information, up to a point $\underline{q}$.

If $\Psi(q) \ge x_{LT}$, information is better in expectation than shutting down learning and receiving a moderate payoff. Pursuing $x > x_{LT}$ both increases flow utility and the weight on the revelation of favorable information. For high q, because information is favorable, $x < x_{LT}$ may counterintuitively be preferred to x_{LT} after a point $\overline{q}$.

The solution to this *unconstrained* problem is shown in the left panel of Figure 3 in red. The vertical axis in this figure represents the *utility space* of the leader. We let $\underline{u} < 0$ so that $\Psi(q) < 0$ for low q. The arrows reflect the direction of the leader's *preference* in the unconstrained problem, which tends to the learning trap below $\underline{q}$ and extreme policies above $\overline{q}$.[7] Below $\underline{q}$, the leader prefers the learning trap for flow utility and informational purposes; between $\underline{q}$ and $\overline{q}$, prefers a high policy for flow utility purposes; and above $\overline{q}$, prefers any high or low policy to x_{LT} for

[7]Formally, the arrows show the sign of the derivative.

informational purposes.

The right panel transposes the constraint set $\mathrm{NR}(q)$ (shaded) onto the *policy* space. We overlay the leader's preference onto $\mathrm{NR}(q)$ to trace out the leader's solution $x_\ell(q)$ in the constrained problem (bolded in red). This forces the leader to pursue $x < x_{LT}$ for low q.

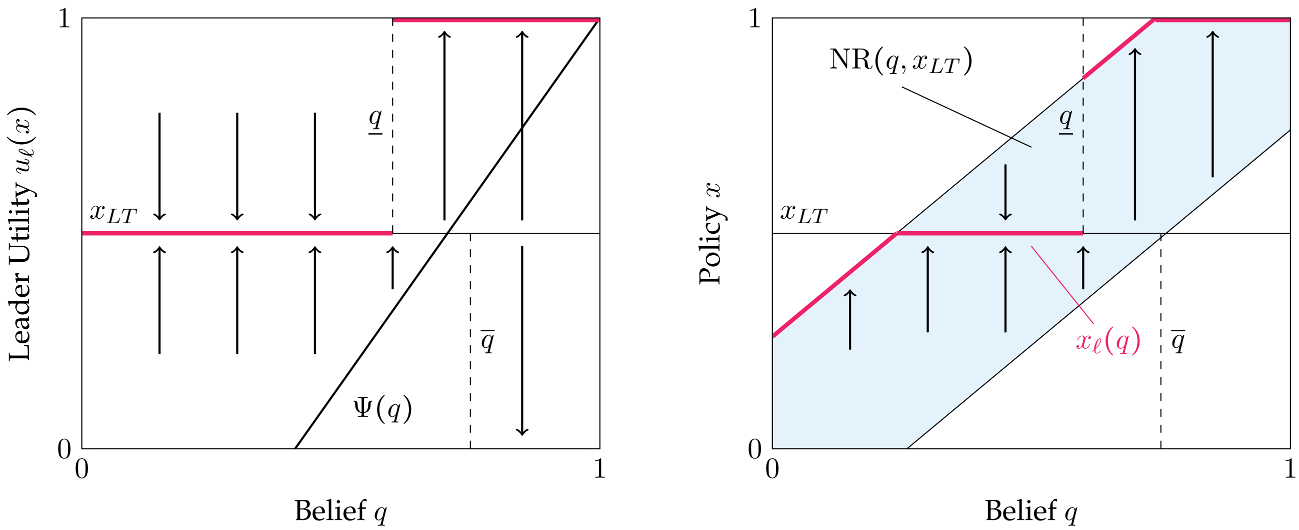

Figure 3: Leader's policy preferences in unconstrained leader's problem (left) vs. leader's constrained solution (right)

□

If the optimal policy for the people is uncertain (intermediate q), leaders moderate policies to suppress information that could constrain them in the future. In fact, the people have a strict incentive to *retain* them, contrasting with dictator games where leaders always keep the people indifferent to overthrow. When leaders are viewed favorably (high q), they experiment with their most preferred policies. When viewed unfavorably (low q), they concede to constituent demands.

The result also yields the following comparative statics.

Proposition 2. *$\underline{q}$ is increasing in δ, decreasing in σ, and decreasing in $\underline{u}$.*

Leaders with high δ value flow utility less, for example, if $(1 - \delta)$ represents a term length. At lower σ, for each x, the probability of information revelation increases. In both cases, the leader is more worried about how her actions may reveal the truth, increasing attraction to learning traps. For $\underline{u}$ low, the value of revealing information is lower at each q. Utilizing our normalization, if

$\underline{u} \equiv u_\ell(\overline{x}(0)) = \overline{x}(0)$, utility is linear. Thus, as $\underline{u}$ decreases, u_ℓ becomes "more concave"; greater aversion to low policies makes the leader more willing to halt learning.

Policy Dynamics Figure 3 provides insights into policy variation when a leader inherits a status quo or faces exogenous shifts in beliefs. Suppose a leader inherits an "out-of-equilibrium" policy $x_0 = 1$ and belief q_0. If q_0 is intermediate, the leader will shift from $x_0 = 1$ to x_{LT}. If q_0 is low, the leader will shift to $\overline{x}(q_0)$, the closest policy to x_{LT} that averts overthrow. If q_0 is high, she will stay at $x = 1$. Thus, despite the fact that $x = 1$ is ideologically favorable to the leader, at intermediate beliefs, she may shift to a less-desirable policy because of its informational features.

Conversely, suppose the leader is already pursuing her optimal policy $x_\ell(q_0)$, but that q_0 unanticipatedly shifts to q_0'.[8] Suppose q_0 is intermediate so that $x_\ell(q_0) = x_{LT}$. If q_0' is slightly above or below q_0, $x_\ell(q_0') = x_{LT}$; the leader's policy is *not sensitive* to small shifts in beliefs. If q_0' is closer to 0, policy shifts towards $x = 0$; the leader is forced to pursue a lower policy by the no-overthrow constraint. If q_0' is closer to 1, the leader's policy swings towards $x = 1$. Thus, beginning at intermediate beliefs, small shocks to beliefs generate no policy variation. Larger shocks favoring the anti-leader state generate shifts to lower policies. Since $x_\ell(q)$ is discontinuous at $\underline{q}$, larger pro-leader shocks generate potentially more dramatic swings to high policies.

Comparison to Short-Lived Leaders How does an infinitely-lived leader's behavior compare to a leader who lives two terms? In a two-period model, a leader would always implement the highest possible policy in the second period, $\overline{x}(q)$. Hence, the value of pursuing x_{LT} would not be x_{LT} for all time, but x_{LT} today and $\overline{x}(q)$ tomorrow. At high beliefs, a leader who values tomorrow's utility would halt learning. This illustrates a fundamental contrast between "short-lived" officials who face salient reelection threats and entrenched incumbents or dictators. Short-lived leaders face replacement if viewed unfavorably, and only experiment when they need to save their chances of retention. Infinitely-lived politicians experiment if they are likely to align with constituents' interests, but otherwise favor learning traps.

Comparison to Model with Outside Option Many political agency models study how leaders make decisions when their constituents are uncertain about their competence. If a leader is in-

[8] Section 5 solves the model when shifts are *anticipated*.

competent, constituents can replace her with an outside challenger. Proposition 1 would not hold if the people received a constant outside option upon overthrow. Suppose x_{LT} were preferred to the outside option at $q = 1/2$. Because the people's utility from x_{LT} is identical in both states, at $q = 0$, x_{LT} would still be preferred to the outside option. The leader could then achieve utility at least x_{LT} at $q = 0$, eroding the need to implement x_{LT} at higher beliefs. The model is thus particularly relevant when constituents are uncertain about a *policy fundamental*, since their replacement decisions are a function of their information.

3.2 Model with Preference for Learning

We have so far assumed the people value only current-period payoffs. However, if they value future payoffs — and thus learning — can they erode leaders' willingness to minimize information? This section establishes that the fundamental insights of the model extend to this general setting: the leader will attempt to *minimize* learning at intermediate beliefs, but non-myopia warps the no-overthrow constraint and generate further nonmonotonicities in leaders' choices. Moreover, it shows that when constituents are patient, a friction like costly experimentation becomes *necessary* to rationalize leaders' choices of learning traps in equilibrium.

The people's discount factor is now $\delta > 0$ and the leader's δ^{ℓ}. To capture experimentation frictions, we allow the people bear adjustment costs $\kappa|x_t - x_{t-1}|$.[9] Flow utility is $u_p(y_t, x_t, x_{t-1}) = y_t - \kappa|x_t - x_{t-1}|$. We assume $f_g'(x_{LT}) > (1-\delta)\kappa$, which ensures adjustment is not *so* costly that policy never moves. We calculate Markov Perfect Equilibrium with respect to the prior q and yesterday's policy x'. $x_p(q, x')$ represents the people's optimal policy conditional on overthrow. In the benchmark, the leader was able to halt learning without any adjustment costs. However, patient constituents may overthrow a leader who halts learning, meaning a leader may not be able to implement a learning trap without some form of costly experimentation.

Solution to People's Problem The solution to this more general version of the model is technical. We preview the results here while delegating proofs to the Supporting Information Appendix.

[9] Forcing the leader to bear these costs further strengthens the result.

First, we solve the people's new problem. Their optimal policy $x_p(q, x')$ solves:

$$W(q,x') = \max_{x\in[0,1]} \underbrace{(1-\delta)\big(\mathbb{E}[y|x] - \kappa|x-x'|\big)}_{\text{Flow utility}} + \delta\left(\underbrace{\frac{\Delta(x)}{2\sigma}\big(qW(1,x) + (1-q)W(0,x)\big)}_{\text{Truth revealed}} + \underbrace{\Big(1-\frac{\Delta(x)}{2\sigma}\Big)W(q,x)}_{\text{Truth not revealed}} \right) \tag{6}$$

The first term represents flow utility. The second set of terms represents continuation value, which depends on the information policies reveal. Proposition 3 in the Supporting Information Appendix characterizes $x_p(q, x')$, which is displayed in Figure 4 and can be described as follows. Suppose the status quo is x_{LT}. For intermediate q, the people face a tradeoff: extreme x generates information — which is incredibly valuable. But because they do not know whether high or low x is optimal, they may incur excessive costs by experimenting in one direction, learning optimal policy is in the other, and "backtracking." The solution is discontinuous around $q = \frac{1}{2}$, where experimentation with high *or* low x is risky. Thus, they pursue policies closer to x_{LT}, although never x_{LT} itself, since this would preclude learning. At extremal q, the people are relatively certain about the state, so there is no tradeoff. $x_p(q, x')$ is bolded below, with x' calibrated to x_{LT}.

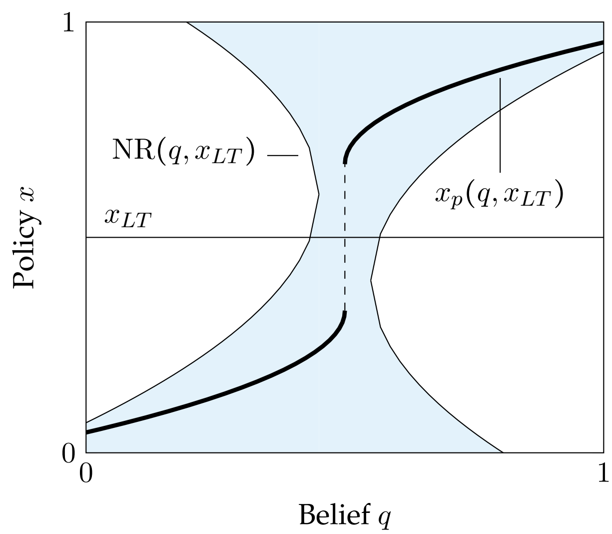

Figure 4: Optimal policy $x_p(q, x_{LT})$ and no-overthrow constraint $\mathrm{NR}(q, x_{LT})$ when people value learning

The figure additionally graphs $\mathrm{NR}(q, x')$, the set of policies that allow the leader to avoid overthrow at belief q and status quo x'. In the baseline, fixing (q, x'), $\mathrm{NR}(q, x')$ was a closed interval. However, since extreme policies are preferred to x_{LT} due to their informational value, this may no longer be the case when the people have a preference for learning.

The appendix shows that a sufficient condition for x_{LT} to be implemented at some point *in equilibrium* is $2(1-\delta)\kappa \geq f_g'(x_{LT})$. Intuitively, if the people are patient and experimentation is costless, $x_{LT} \notin \mathrm{NR}(q, x')$, and the leader cannot implement x_{LT} in equilibrium. Thus, $(1-\delta)\kappa$ must be sufficiently large to have $x_{LT} \in \mathrm{NR}(q, x')$ at some (q, x') Nevertheless, even if $x_{LT} \notin \mathrm{NR}(q, x')$, the leader can still minimize learning by pursuing the *closest policy* to x_{LT}.

Leader's Problem The leader's problem is similar to that in (4). At $q = 0$ or 1, the leader will pursue the most extreme policy preventing overthrow. The equation describing her optimal solution $x_\ell(q, x')$ is:

$$V(q, x') = \max_{x \in \mathrm{NR}(q, x')} (1-\delta^\ell) u_\ell(x) + \delta^\ell \left(\frac{\Delta(x)}{2\sigma} \Psi(q, x) + \left(1 - \frac{\Delta(x)}{2\sigma}\right) V(q, x) \right) \tag{7}$$

where $\Psi(q, x')$ represents the *expected value* of revealing information. We let $\underline{u}(x')$ describe the leader's expected utility conditional on the anti-leader state realizing given status quo x'. The following Theorem describes $x_\ell(q, x')$.

Theorem 1. *Suppose $\sigma(1-\delta^\ell)$ is small or $\underline{u}(x')$ is sufficiently low. There exist thresholds $\underline{q} < \overline{q} \in (0, 1)$ such that:*

1. *If $q \leq \underline{q}$, the leader plays the* closest policy *to the learning trap (which may be* greater *or* less *than x_{LT}).*
2. *If $q \in [\underline{q}, \overline{q}]$, the leader plays either the highest policy below the learning trap or her highest feasible policy.*
3. *If $q > \overline{q}$, the leader either plays her highest feasible policy or lowest feasible policy.*

The central forces of the model are identical to the baseline: leaders prefer learning traps at intermediate and low beliefs and their preferred policy otherwise. The key difference in equilibrium behavior is driven by the no-overthrow constraint. At some beliefs, the leader may be unable to pursue x_{LT}, but may be allowed to pursue high policies because of their informational value. Conditional on being forced to reveal some information, leaders prefer these high policies, as shown in Figure 5.

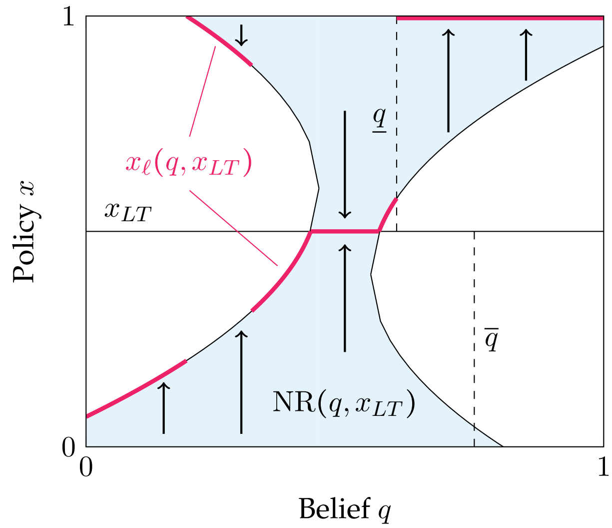

Figure 5: Leader's constrained solution when people value learning

3.3 Additional Extensions

We now discuss additional extensions to the benchmark model: information asymmetry, alternative noise specifications, multiple states of the world, and heterogeneous (or ambiguous) beliefs. Proofs are in the Supporting Information Appendix.

Information Asymmetry The baseline model assumes the leader and people possess symmetric information about the state. However, leaders may have more access to information about policies and their effects (e.g., through the public service). How would asymmetric information change the model's insights?

If the leader received a *private* signal about the state, she would use it to assess the value of revealing information. Her actions would be constrained by *public* information via the no-overthrow constraint.[10] Suppose we begin at a belief where the leader pursues x_{LT}. If the signal weakly favors her, the leader may still want to pursue x_{LT}. If the signal favors the anti-leader state, she will surely want to pursue x_{LT}. The people cannot infer her signal from her actions, and

[10]If the leader and the people simply held different (subjective) beliefs, the people's beliefs would affect the no-overthrow constraint, while the leader's would affect the value of information.

the model's predictions remain the same. However, if a pro-leader signal changes the leader's choice, her actions reveal her signal, meaning playing the learning trap leads to a drop in q.

Private information can thus make the leader *worse off*. Consequently, if a long-lived leader could *commit* to setting up an institution that monitored the effects of experimental policies, she may choose not to.

Model with Non-Uniform Noise Are the insights of the model robust to alternative distributions of the noise shock ϵ_t — such as a normal distribution? Under a technical condition, the result generalizes. Suppose $\epsilon_t \sim p$, where $p(\cdot)$ is continuous, symmetric around 0, and single-peaked at 0. Denote the distribution of posteriors q_{t+1} given a prior q_t and policy x_t as $\tau(q_{t+1}|q_t, x_t)$. Suppose x_t' is farther from x_{LT} than some other policy x_t. The variance of posteriors $\tau(q_{t+1}|q_t, x_t')$ is larger than $\tau(q_{t+1}|q_t, x_t)$ but has the same mean (the prior q_t). Policies farther from x_{LT} reveal more information. The leader's problem can be expressed as:

$$V(q) = \max_{x \in \mathrm{NR}(q)} \underbrace{(1-\delta)u_\ell(x)}_{\text{Flow utility}} + \underbrace{\delta \int V(q')\tau(q'|q,x)}_{\text{Value of information}} \tag{8}$$

The continuation value $V(q')$ is integrated over the posteriors $\tau(q'|q,x)$. When $x = x_{LT}$, $\tau(q'|q,x)$ places mass 1 on the prior; as x grows extreme, the variance of posteriors increases. There is an aversion to information if $V(q)$ is concave, which holds if the *graph* of $\mathrm{NR}(q)$ is convex, as in the baseline (Figure 2). The following proposition characterizes $x_\ell(q)$.

Proposition 4. *Suppose the graph of* $\mathrm{NR}(q)$ *is convex. There exists a threshold* $\underline{q} \in (0,1]$ *such that:*

1. *if* $q \leq \underline{q}$, $x_\ell(q) = \overline{x}(q)$. *The leader plays the highest policy preventing overthrow.*
2. *If* $q > \underline{q}$, *the leader implements* $x_\ell(q) \in [\min\{x_{LT}, \underline{x}(q)\}, \overline{x}(q)]$. *Moreover, for* δ *high,* $x_\ell(q) < \overline{x}(q)$. *I.e., she implements a policy closer to the learning trap than her maximal policy* $\overline{x}(q)$.

Multiple States of the World The benchmark model considers an "anti-leader" and "pro-leader" state. The Supporting Information Appendix shows that the model's intuitions generalize if there is a third "middle" state, where the optimal policy for constituents is moderate.

While leaders cannot totally shut down information between all three states, they can pool information "pairwise." For example, they can implement a policy that separates the "anti-leader" state but keeps constituents uncertain between the "pro-leader" and "middle" states. Players will eventually learn that either the "anti-leader" state is true, or that one of the "pro-leader" and "middle" states is true, but not which. The appendix shows that the leader pools the "anti-leader" with either the "pro-leader" or "middle" state, but never separates the "anti-leader" state.

The model with two states can represent the terminal history of a model with more states. Suppose the leader pools the "anti-leader" and "pro-leader" states while separating the "middle" state. Then, either the "middle" state realizes or constituents learn that the "middle" state does not hold — in which case the model returns to the benchmark with a pro and anti-leader state.

Model with Ambiguity Aversion We have assumed all players have a common prior q in the pro-leader state. However, the people may be diverse in their beliefs, or the leader may be uncertain about where constituents' subjective beliefs lie. How does this affect leaders' behavior?

We study the leader's behavior under *ambiguity aversion*: instead of behaving as if the people's belief in the pro-leader state is q, she believes their beliefs lie in $\mathcal{Q} \subseteq [0,1]$. The leader behaves as a "maxmin expected utility" maximizer in the sense of Gilboa and Schmeidler (1989), meaning she chooses each policy anticipating the corresponding "worst case scenario." This behavior can also reflect the leader's aggregation of a diverse set of beliefs that requires appeasing the "most pessimistic" of her constituents.

We start with an example evaluating the leader's continuation utility. Suppose the leader can pursue $x = 0$ or $x = 1$ today and avert overthrow. The prior q moves to the posterior q', which is always either 0, 1, or q. The leader evaluates the problem as:

$$\max_{x \in \{0,1\}} \min_{q' \in \mathcal{Q} \cup \{0,1\}} (1-\delta)u_\ell(x) + \delta V(q') = \max_{x \in \{0,1\}} (1-\delta)u_\ell(x) + \delta \underline{u} \tag{9}$$

She thus evaluates information revelation as if the anti-leader state materializes. Returning to the original problem, suppose the leader believes q lies in a compact set $\mathcal{Q} \subseteq [0,1]$ with $\mathcal{Q} \cap (0,1) \neq \varnothing$.

Her problem is:

$$V(\mathcal{Q}) = \max_{x\in[0,1]} \min_{q\in\mathcal{Q}} \begin{cases} (1-\delta)u_\ell(x) + \delta \min_{q'\in\mathcal{Q}} V(\mathcal{Q}) & x \in \mathrm{NR}(q), x = x_{LT} \\ (1-\delta)u_\ell(x) + \delta \min_{q'\in\mathcal{Q}\cup\{0,1\}} V(q') & x \in \mathrm{NR}(q), x \neq x_{LT} \\ -L & x \notin \mathrm{NR}(q) \end{cases} \tag{10}$$

If she pursues x_{LT}, no information is revealed; her continuation value is $\min_{q'\in\mathcal{Q}} V(\mathcal{Q})$; if $x_{LT} \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$ today, it is also implementable tomorrow. If she implements $x \neq x_{LT}$, information is revealed, which she evaluates as in (9). There may also exist no policies that are in $\mathrm{NR}(q)$ for all q, in which case the leader may be deposed.

The following result shows that, where possible, the leader will implement the learning trap, since she evaluates information as if it favors the anti-leader state.

Proposition 5. *Let $\mathcal{Q} \subseteq [0,1]$ be a compact set of constituent beliefs such that $\mathcal{Q} \cap (0,1) \neq \varnothing$. Suppose the leader operates as a maxmin expected utility maximizer and δ is large. Let $x_\ell(\mathcal{Q})$ be the leader's optimal policy conditional on believing the people's belief q lies in $\mathcal{Q}$. Then:*

- *if $x_{LT} \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = x_{LT}$;*
- *if $x_{LT} \notin \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$ but there exists x such that $x \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q})$ is the highest policy that prevents overthrow;*
- *if there does not exist $x \in [0,1]$ such that $x \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = 1$.*

4 Application: The 1861 Reforms as a Learning Trap in Imperial Russia

This section illustrates how the model's informational insights help us understand long-lived regimes' policy choices. I study the reforms ending Serfdom in Imperial Russia, showing how the Tsar's pursuit of a "middle-ground" economic system that inefficiently mixed communal decisionmaking with elements of a market economy resembled a learning trap. I map the Russian case to the model and argue that this "learning trap" was a response to uncertainty about the alignment of the Tsar's preferred policy with everyday Russians' interests. I discuss alternative equilibria and explanations in the "Additional Case Study" material of the Supporting Information Appendix.

4.1 Background

Before 1861, most Russian peasants provided services or payments to the state or a noble in exchange for use of arable land. Peasants working noble land were "serfs," and the system closely resembled slavery.[11] Seigneurs oversaw a "commune" of households in charge of repartitioning land, solving disputes, and managing resources. Many peasants possessed limited control over production decisions, had few rights, and could be tied to land in the event of sale (Vasudevan (1988, 209), Nafziger (2010, 382), Nafziger (2016, 775-776) Dennison (2014, 252-257)). After Russia's defeat in the 1853 Crimean War, many Russians maintained that the Tsar lost the war due to economic backwardness; discourse about alternatives to Serfdom began circulating (Starr, 2015, 57). Some nobles argued that allowing serfs to privatize communal land would improve economic growth (Starr, 2015, 53-63). Others wanted nobles to own land, with peasants renting and working it "on the basis of free competition" (Khristoforov, 2009, 65).

Regime's Preferred Policy The Russian regime, headed by Tsar Alexander II, likely preferred a liberal path to economic modernization that resembled Western Europe — characterized by household private property, individual rights to output, and freedom for labor to move to cities — due to its implications for development, fiscal capacity, and the military. Russia's widening military gap with its neighbors was attributed to Serfdom's role in stifling the emergence of industry (Zenkovsky (1961, 284), Emmons (1966, 48-50)) and the State's ability to conscript the peasantry (Moon, 2014, 53-54). Endowing peasants with private property could allow labor to flow to cities — generating industrial growth — and mitigate frictions that made it difficult to recruit peasants from nobles' land. A liberal system could accelerate the development of state institutions through its fiscal effects (Dennison, 2023, 642). Under Serfdom, tax collection was inefficiently outsourced to nobles who siphoned surpluses (Dennison, 2020, 2023). Liberal reform could weaken the nobility's bite, increasing fiscal revenue (Moon, 2014, 33). Reformers hoped to use fiscal revenues to

[11] Peasants rarely owned land, but some owned small private parcels before and after 1861. Nafziger (2012) shows how peasants on State-owned lands possessed some more individual property rights than private serfs.

increase state capacity and expand corporate forms and military ventures (Dennison, 2023, 642). In an ideal world, liberal reform would be *aligned* with constituents' interests as well: it could increase worker mobility and productivity, generate growth, protect Russians during war, and increase quality of life (Pereira, 1980, 108).

Model Mapping We view Russia through our model as follows. One extreme policy is liberal reform, while the other is the status quo of Serfdom. In the pro-leader state of the world, the government's preferred policy — liberal reform — would be aligned with the people's interests — a combination of the peasantry and nobility. In the anti-leader state of the world, the government's interests would be unaligned with constituents, who would force them to maintain Serfdom or pursue non-liberal approaches to development. Many Russians *did* believe that elements of Serfdom, like the commune, offered benefits over liberal reform. Slavophilic ideologues romanticized the peasant as a self-sufficient "bearer of his own culture" (Khristoforov, 2009, 62) and argued the commune should be left untouched. Some believed private property could lead to a countryside marred with disorder, requiring seigneurial intervention to prevent exploitation of commons (Pravilova, 2014). Others contended that peasants were not aware of issues with the pre-1861 system or preferred communal land (Field, 1976; Kingston-Mann, 1991). A minister of internal affairs contrasted Russia's "uninterrupted tranquility" under Serfdom with the "[i]nternal discord and revolts" plaguing liberal Europe (Polunov et al., 2015, 87). The Russian countryside was not beset by economic crisis (Moon, 2014, 19-22), and nothing had "challenged the basic structure of local institutions" (Starr, 2015, 52).

At the eve of the 1860s, these debates imply *uncertainty* over whether liberal reform aligned with public interest. The "beginning" of the model can be thought of as a "shock to beliefs" triggered by the Crimean War that shifted opinion towards the pro-leader state. If we interpret the pro and anti-leader states as representing models of the world, we can view the Crimean War as triggering the "introduction" of a pro-leader/liberal reform model.

4.2 Learning Trap, 1861-1905

Government's Fears of Information Revelation How did the government respond to uncertainty about whether liberal reform would be aligned with its constituents' interests? The Tsar understood that his *policy* choices could reveal *information* about the effects of policies that could potentially *constrain* him in the future if liberal reform backfired. Thus, the government viewed "communal land tenure and mobility restrictions [as] necessary precautions" in the face of uncertainty (Dennison, 2014, 62). Endowing peasants with private property *could* lead to alienation of agriculture, limiting the government's ability to collect revenue and stressing state capacity (Dennison, 2023, 644). External validity against the appropriateness of "Western" policies included "the rise of an industrial proletariat, ruthless competition, unemployment, and other evils of industrial capitalism" (Polunov et al., 2015, 61). One of Alexander II's committees concluded that "the Order of the State might be shaken" if they did *not* tie peasants to land (Field, 1976, 75); mixing elements of communalism into the reforms would thus tide these *informational fears*.

Terms of Emancipation Thus, while the Tsar's emancipation reforms beginning in 1861 freed peasants from the coercive status quo of seigneur-peasant agriculture and broadened civil rights (Dennison, 2014), they did *not* remove the peasant commune and used it to constrain household labor mobility and capacity to acquire private property. Nobles were compensated by the State for relinquishing land that was redistributed between members of communes, financed by redemption payments made by serfs over 49 years. After payments were made, ex-serfs would privately own their allotted land but were otherwise *communally* responsible for redemption and tax payments. Households could not legally sell, lease, alienate, or use land as collateral *until* redemption payments were complete, restricting freedom to abandon agriculture (Dower and Markevich (2019, 234), Nafziger (2012, 5-6), Nafziger (2016, 776), (Dennison, 2014, 259)).

Intermediate Policy The 1861 reforms mixed communal Serfdom with liberal institutions. An external seigneur no longer regulated economic decisions. Peasants could privatize communal land by paying off redemption obligations (Nafziger, 2010, 383). Peasants owned houses, gardens, and rights to output (Kingston-Mann, 1991, 35). Communes (and sometimes households) even engaged in market land transactions with formal credit backing (Nafziger, 2010, 384), although redemption payments limited this activity. However, the persistence of the commune and its codification by the State clashed with individual economic incentives. Land allotments, repartitions, and redemption obligations were enacted communally, sometimes against households' wills. Peasants owned scattered strips of land, at best allowing for diversification, at worst increasing travel times and disincentivizing technological experimentation, slowing agricultural productivity growth (Williams (2013, 52-53, 65-66), Dower and Markevich (2019, 243)).

Mobility Restrictions The commune restricted labor mobility and supply by controlling the issuance of documents to pursue outside work (Nafziger, 2012, 7). Because households were communally responsible for redemption payments, those with larger obligations were incentivized to enforce mobility restrictions and free-ride on others. Nafziger (2010) notes: "Income-maximizing peasant households may have been unable to freely allocate resources between agricultural production and non-agricultural activities if they were subject to collective decision-making...By potentially inhibiting land transfers...and restricting the allocation of labor outside the village, the communal system may have introduced wedges between the shadow and market values for these factors of production" (384). These frictions stifled the flow of labor to cities, dampening the emergence of an industrial sector and contradicting one of the original motives for reform.[12]

Implications Many contemporaries felt the Tsar's reforms did not go far enough. In practice,"[f]ew surpluses could be wrung from peasants who were already obligated to repay the nobility for their freedom and their land and who continued to face obstacles to the most efficient allocation of their resources" (Dennison, 2023, 644-5). Coupled with tepid industrial growth and limits to peasant economic activities, why would the State have pursued this middle ground? The model provides an answer via the lens of a learning trap: the Tsar feared information revelation could strain the government if unsuccessful reform revealed its preferences were unaligned with the people. An intermediate system ending many of Serfdom's abuses but still tying peasants to agricultural land could mitigate these informational threats.

Evidence from the ensuing decades suggests that the post-1861 reforms halted elites' and the State's ability to *learn* whether liberal reform would have been better. Despite the commune's apparent discouragement of technological adoption, when innovation did occur, it *spread* more quickly in districts with greater communal intensity (Kingston-Mann, 1991). While many in the government still believed in the pursuit of a liberal, Western path, one source cites a government

[12]Indeed, Nafziger (2010) argues that weaker communal structures allowed Moscow to develop stronger industry.

official,"an enthusiastic critic of the commune" (49), who conceded that the commune's ability to provide mutual aid during emergencies was a benefit that could not be realized under private property. Decades later, officials contended that limited cases where peasants were permitted to sell land caused exploitation of peasant decision-making (Wcislo, 2014, 89), casting doubt about the appropriateness of markets that would accompany liberal reform.

The *general equilibrium* effects of reform likely differed from the partial equilibrium experience of individual villages toying with private incentives. Regional heterogeneity in the enforcement of communal institutions allowed some households to observe the effects of private incentives, accumulating wealth by participating in rental markets and non-agricultural labor (Nafziger, 2010, 385). Nevertheless, the contemporary remarks above suggest high-level uncertainty about whether liberal reform was surely better, even decades after 1861.

Not only did the Tsar's government hold power for decades, but except for a spike in disturbances right after 1861, reported peasant discontent remained low in the following years (Finkel et al., 2015, 1000). In this sense, like a learning trap, the Tsar's intermediate policy was successful in curbing threats of overthrow. However, beyond stifling Russia's industrial sector and potentially slowing agricultural adoption in the medium term, Markevich and Zhuravskaya (2018) go as far as to argue that this system dampened Russian economic growth into the 20th century. [13]

4.3 Other Equilibria and Interpretations for Learning Trap

While the model predicts that the Tsar would pursue a learning trap at intermediate beliefs in the pro-leader state, it also predicts that at extreme beliefs, the government would experiment with liberal reform. The first part of the online "Additional Case Study" material argues this was the case: after shocks to beliefs in 1905 favoring liberal reform, the government experimented with private property under Stolypin.

Second, the historical literature centers two explanations for the Tsar's middle ground: bargaining with the nobility and low state capacity. The second part of the online "Additional Case Study" material argues that while these factors undoubtedly shaped the reforms, the diminishing

[13]Some historians point to redemption payments as the first-order inefficiency impeding growth. Interestingly, Gerschenkron (1962) suggests the commune *worsened* the bite of redemption payments. Households with large debt would have an incentive to free-ride on other commune members, strengthening enforcement of mobility restrictions and distorting production incentives, affecting households' very ability to make payments.

power of the elites, similarities in the experiences of ex-serfs to peasants on state-owned lands, and lack of policy variation over decades cast doubt on whether these forces alone explain the intermediacy of the emancipation reforms. The *learning trap* interpretation helps resolve these puzzles via a *complementary* informational mechanism.

5 Application: Anticipated Information Revelation and Political Contagion

Can the model's insights explain why certain regions of the world experience more policy experimentation? I extend the baseline model to allow for the leader and people to also learn about the state of the world from external sources. I show that this increases experimentation from a direct channel (exogenous information) and an indirect channel (leaders experiment more in response).

This insight allows us to understand the interplay of policy contagion — where information about the effects of policies spreads from one polity to another — and political fragmentation. I interpret 19th century Europe as a region where the probability of external information revelation was high and China as a region where it was low, and argue that the model can explain why European nations experimented with indirect rule — experiencing high political turnover in the process — while China did not.

5.1 Information Revelation

Information revelation may be *anticipated* when leaders are consciously aware of external sources of inference. Officials in regions plagued by earthquakes or hurricanes may foresee that these disasters will eventually occur and reveal information about disaster relief policies. Or, groups of countries with similar institutions learn policy fundamentals from their neighbors' experiments.

I model anticipated information shocks as follows. Let $\eta \in (0,1)$ be small. With probability η at the end of every period, the true state of the world is exogenously revealed. The leader's problem can be written as:

$$\max_{x \in \mathrm{NR}(q)} \frac{2\sigma(1-\delta)u_\ell(x) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)\Psi(q)}{2\sigma(1-\delta) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)} \tag{11}$$

When $x = x_{LT}$, the leader's value is now

$$\frac{(1-\delta)x_{LT} + \delta\eta\Psi(q)}{(1-\delta) + \delta\eta}$$

instead of just x_{LT}. x_{LT} still *minimizes* learning, but is no longer a "safe option" that totally *halts* learning. The following proposition reflects that because she can no longer *fully* control information, she may take extreme actions to enjoy more flow utility.

Proposition 6. *$\underline{q}$ is decreasing in η. The leader plays the learning trap for fewer values of q.*

Anticipated information revelation improves learning compared to the baseline. While there is more information revealed mechanically via η, because $\underline{q}$ decreases, the leader experiments with her preferred policy for a larger range of q values, meaning learning also increases through an *endogenous* channel. This means that fragmented regions plagued by policy contagion will see more governments making concessions to their people. Or, insofar as low x is shorthand for increased external threats of overthrow, leaders' terms will be shorter, corresponding to more frequent political turnover.

5.2 Political Fragmentation in Europe and China

The model's prediction — politically fragmented regions of the world exhibit larger policy variation, more policy experimentation, and more frequent political turnover — is consistent with stylized historical facts differentiating Western Europe and China. It explains how the spread of constitutional monarchies and representative democracies in fragmented Western Europe contrasted with the consistent stability of consolidated Imperial China. Because systems of indirect rule were not widely experimented with by Eurasian states until the 1700s, we view our "uncertain policy" as the optimal degree of democracy or autocracy. The policy space corresponds to direct rule (e.g. monarchy) on one end and indirect rule (e.g. republican democracy) on the other. Constitutional monarchies can be thought of as situations where autocrats begrudgingly cede power to the people but remain in office.

Europe was populated by dozens of sovereign states around 1800 and possessed rich sources of policy information, i.e. high η (Dincecco and Wang, 2018). England's "Glorious Revolution" at the end of the 17th century marked Europe's first constitutional monarchy (Kurian, 2011) and one of the first sources of data on indirect rule. After the French Revolution, rulers began to worry about citizens' desires for indirect rule (Evans and Von Strandmann, 2002, 10) as demands for freedom of speech, press, and disposal of private property grew (Sperber, 2005, 66-67). By the first half of the 19th century, both Britain (constitutional monarchy) and America (representative democracy) were providing information on the efficacy of indirect rule. As these informational pressures strengthened, by 1850, most European states shifted to some form of government partially controlled by an elected legislature (Tilly, 1989). Some governments peacefully ceded power, while others had it wrested from them in an "Age of Revolutions." External information revelation could thus be thought to force governments to experiment with indirect rule, increasing learning and, as the model would predict, in some cases generating political upheaval.

In contrast, China was ruled by a unified State uninterrupted for a millennium. No states shared China's scale or political structure, suggesting low η in 1800. China possessed relatively integrated economic institutions that allowed its central government to oversee trade and a tightly-managed bureaucracy (Rosenthal and Wong, 2011). The state's Confucian ideology centered political stability, contrasting with the "disruptive progress" of Western Europe. These forces arguably prevented the emergence of ideologies promoting indirect rule, which was only strengthened by the lack of external information (Mokyr, 2016, ch. 16). China's high (although stagnant) economic standards in the late 1700s evidence a degree of policy *moderation* that was not seen in Western Europe (Pomeranz, 2021). Rare threats of overthrow from regional elites seemingly strengthened — rather than eroded — the central government's power (Dincecco and Wang, 2018).

The "high η" environment of Western Europe in the 18th and 19th centuries was accompanied by greater policy variation, concessions to liberal demands, and political transitions. The "low η" environment of China experienced little variation and minimal turnover. This prediction is consistent with Mokyr (2016), who argues that "there were repressive and reactionary regimes galore in Europe, but the interstate competitiveness constrained their ability to enforce a specific orthodoxy" (317). This prediction also complements a literature arguing that Europe's political fragmentation drove the Great Divergence by fostering competition over technology (Diamond, 2005), intellectual networks (Mokyr, 2016), and institutional innovation (Cox, 2017).

6 Additional Applications

Finally, I provide additional examples of how learning traps can help understand leaders' policy decisions. Take Soviet Premier Leonid Brezhnev's "developed socialism," marketed as a transition phase between a state's development of socialist infrastructure and convergence to a communist society (Sandle, 2002, pp. 165-166). Despite his predecessor Khrushchev's commitment to achieving full communism by 1980, Brezhnev refused to adopt the necessary reforms to achieve this goal, pursuing counter-reform to his own 1965 planning and production policies, and making it hard to argue that growth would have been higher *or* lower without the counter-reforms (Harrison, 2002, pp. 57-58). The model provides insights into Brezhnev's dilemma: he was unsure whether communism would work — hence whether his preferred policy was aligned with constituents' — and pushed an inefficient middle ground that prevented constituents from learning the effects of "true communism."

A second example comes in Franklin D. Roosevelt's refusal to include national health insurance in the 1935 Social Security Act. FDR dropped it "at the last minute" (Quadagno, 2006, p.23), and while he convened a committee on medical care that recommended the expansion of public health programs, he instructed "organizers to be circumspect about any possible next steps" (Hoffman, 2012, p. 29), refused to endorse a Senate call to fund state-level insurance, and permitted the proliferation of private insurance. Despite the opposition of the physician lobby, FDR had the power to expand government intervention in healthcare, and his actions are inconsistent with his belief in healthcare as a human right (Hoffman, 2012, p. 36). One way to read FDR's actions is that he believed the public was uncertain whether government-run health programs would actually be aligned with their broad interests, and that failure could curtail future expansions of the welfare state more broadly.

Finally, learning traps can also be applied to "limited" (rather than "total") war. The United States' extended involvement in Afghanistan in the 2000s and 2010s was characterized by deliberate limitations in troop deployment that hampered efforts "to gain sufficient military victories to force a political settlement" Griffin (2014). This limited involvement accompanied widespread public uncertainty about whether there was hope of salvaging Afghan political stability through external intervention (Biddle et al., 2010). Viewed through the model, a leader who is uncertain about her country's ability to win an existing conflict may moderate her actions. Full-scale involvement reveals surely whether victory is possible (which ends in sure defeat or loss), while full-scale withdrawal usually signals a war is lost. Limited war provides cover: any gains or losses can be attributed to partial involvement. Yet, just as learning traps must be perpetually maintained to preserve uncertainty, a leader may need to sustain a limited war over an extended period to uphold these reputational effects.

7 Conclusion

This paper studies a repeated model where a leader is uncertain about whether her preferred policy benefits her constituents. Constituents learn about the effects of policies from her policy choices. Thus, the leader's choices generate *information* affecting constituents' beliefs about their alignment with her policy preferences, and thus what she can implement. When beliefs are intermediate — so the optimal policy for constituents is unknown — the leader pursues a spatially intermediate *learning trap* policy — shutting down information revelation when it would be most useful. When beliefs favor the leader, she experiments with her preferred policy, and when they disfavor the leader, she concedes to constituents.

These predictions contrast with the behavior of short-lived leaders who "gamble for resurrection," shutting down learning when viewed favorably and experimenting when viewed unfavorably. By contrast, long-lived leaders in persistent dictatorships or entrenched incumbencies halt learning when the effects of policies are unknown, but may experiment if viewed favorably. I show how the insights of this repeated model are robust to information asymmetries, additional preferences for constituent learning, multiple states of the world, ambiguity about constituents' beliefs, and external information revelation.

I then use a case study of the end of Serfdom in Russia to illustrate how the informational mechanisms of the model plausibly played a role in the policy choices of Tsar Alexander II. In the late 1850s, the Tsar wanted to modernize the Russian economy by pursuing Western-style economic reform. However, his government was unsure whether this preferred policy of "liberal reform" would actually work in Russia due to its cultural and institutional differences with the West. As a result, the government implemented an intermediate policy that mixed elements of communal and liberal institutions, creating an inefficient "middle ground" that distorted labor and production decisions, restricting economic, fiscal, and industrial growth. However, I argue that the choice of this middle ground can be thought of as a learning trap: by preventing constituents from being able to discern the effects of different policies, the government was able to halt learning and keep itself in power for decades.

I augment the model with external information revelation to study how policy contagion can affect leaders' actions, showing that revelation of information from externally valid sources — such as in politically fragmented regions — increases leaders' willingness to experiment. I show how the model can thus explain why Europe experienced political upheaval and convergence to systems of indirect rule in the 19th century while China did not, complementing "fragmentation" hypotheses of the Great Divergence. Furthermore, I show how learning traps can also be applied to understand Brezhnev's refusal to adopt "full communism" in the USSR, FDR's unwillingness to pursue healthcare reform, and why polities like the US may wage limited wars.

Long-lived leaders halt learning when their constituents are uncertain about the optimal policy, and learning would be most useful. While term limits may provide an appealing solution, the "gambling for resurrection" literature suggests this creates the opposite problem: leaders only experiment when likely to be *misaligned* with constituents and otherwise halt learning. Future work will thus investigate accountability mechanisms that encourage efficient policy experimentation in both repeated and term-limited settings.

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Online Supporting Information Appendix for “Learning Traps”

- **Proofs of Results with Patient People** A2
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Proofs of Results with Patient People

Proof of Solution to People's Repeated Problem

The following proposition characterizes the solution to the people's problem when there is no preference for learning. We normalize flow utility to $(1-\delta)\big(\mathbb{E}[y|x]-\kappa|x-x'|\big)$ and define $k = \kappa(1-\delta)$. The people's Bellman equation is $W(q,x')$ and its solution $x_p(q,x')$:

$$W(q,x') = \max_{x\in[0,1]} (1-\delta)\mathbb{E}[y|x] - k|x-x'| \\ + \delta\left(\frac{\Delta(x)}{2\sigma}\big(qW(1,x) + (1-q)W(0,x)\big) + \big(1 - \frac{\Delta(x)}{2\sigma}\big)W(q,x)\right).$$

We define $y_{LT} \equiv f_g(x_{LT}) = f_b(x_{LT})$.

Proposition 3. *$x_p(q,x')$ is also the solution to the following equation:*

$$\max_{x\in[0,1]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q,x)}{2\sigma(1-\delta)+\delta\Delta(x)} - k|x-x'|$$

where $\Phi(q,x) = qW(1,x) + (1-q)W(0,x)$. Moreover, if $k < f_g'(x_{LT}) < 2k$, there exists $c > 0$ and a set $LT \subseteq [0,1]\times X$ with $LT \ni (1/2, x_{LT})$ and nonempty interior such that for all $(q,x') \in LT$, $W(q,x') \leq y_{LT} + c \leq W(0,x'), W(1,x')$.

The proof for this result is contained in Lemmata 1-5.

Lemma 1. *Suppose x^* is a solution to the functional equation defining $W(q,x')$. Then, x^* is also a solution to the functional equation defining $W(q,x^*(q,x'))$.*

Proof. Suppose by contradiction that x^{**} solves the functional equation associated with $W(q,x^*)$ but not $W(q,x')$. Since both x^* and x^{**} are always feasible, this implies

$$(1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**}-x^*| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1,x^{**}) + (1-q)W(0,x^{**})) + \delta(1 - \frac{\Delta(x^{**})}{2\sigma}W(q,x^{**}) \\ > (1-\delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}(qW(1,x^*) + (1-q)W(0,x^*)) + \delta(1 - \frac{\Delta(x^*)}{2\sigma})W(q,x^*),$$

which also implies

$$
\begin{aligned}
&(1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**}-x^*| - k|x^*-x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1,x^{**}) + (1-q)W(0,x^{**})) \\
&+ \delta(1-\frac{\Delta(x^{**})}{2\sigma})W(q,x^{**}) \\
&> (1-\delta)\mathbb{E}[y|x^*] - k|x^*-x'| + \frac{\delta\Delta(x^*)}{2\sigma}(qW(1,x^*) + (1-q)W(0,x^*)) + \delta(1-\frac{\Delta(x^*)}{2\sigma})W(q,x^*).
\end{aligned}
$$

However, since $k|x^{**}-x'| \leq k|x^{**}-x^*| + k|x^*-x'|$, this implies

$$
\begin{aligned}
&(1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**}-x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1,x^{**}) + (1-q)W(0,x^{**})) + \delta(1-\frac{\Delta(x^{**})}{2\sigma})W(q,x^{**}) \\
&\geq (1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**}-x^*| - k|x^*-x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1,x^{**}) + (1-q)W(0,x^{**})) \\
&+ \delta(1-\frac{\Delta(x^{**})}{2\sigma})W(q,x^{**}) \\
&> (1-\delta)\mathbb{E}[y|x^*] - k|x^*-x'| + \frac{\delta\Delta(x^*)}{2\sigma}(qW(1,x^*) + (1-q)W(0,x^*)) + \delta(1-\frac{\Delta(x^*)}{2\sigma})W(q,x^*),
\end{aligned}
$$

contradicting that x^* solved the functional equation associated with $W(q,x')$. □

Lemma 2. *There exist* $0 \leq \underline{x} < x_{LT} < \overline{x} < 1$ *such that*

$$
W(1,x') = \begin{cases} f_g(\overline{x}) - k(\overline{x}-x') & x' \leq \overline{x} \\ f_g(x') & x' > \overline{x} \end{cases} \qquad W(0,x') = \begin{cases} f_b(\underline{x}) - k(x'-\underline{x}) & x' \geq \underline{x} \\ f_b(x') & x' < \underline{x} \end{cases}
$$

Proof. Suppose $q=1$ and $x' \leq 1$. $W(1,x') = \max_{x\in[0,1]}(1-\delta)f_g(x) + \delta W(1,x) - k|x-x'|$. By Lemma 1, this simplifies to $\max_{x\in[0,1]} f_g(x) - k|x-x'|$. If $f_g(x') \geq k$, this is solved by $\overline{x} = 1-k/2$. Otherwise, x' itself solves the equation. That these points are to the right of x_{LT} follows from $f_g'(x_{LT}) > k$. A similar argument applies for $W(0,x')$. □

Lemma 3. *Suppose* x^* *solves the functional equation defining* $W(q,x')$ *and* $q \in (0,1)$. *Then:*

$$
W(q,x^*) = \frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)}
$$

where $\Phi(q,x) = qW(1,x) + (1-q)W(0,x)$.

Moreover, $W(q,x') = \max_x \frac{2\sigma(1-\delta)\mathbb{E}[y|x]+\delta\Delta(x)\Phi(q,x)}{2\sigma(1-\delta)+\delta\Delta(x)} - k|x-x'|$.

Proof. Since x^* solves the functional equation associated with $W(q,x')$, by the previous lemmata it also solves the one associated with $W(q,x^*)$. Hence:

$$
\begin{aligned}
W(q,x^*) &= (1-\delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q,x^*) + \delta(1-\frac{\Delta(x^*)}{2\sigma})W(q,x^*) \\
&= \frac{(1-\delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q,x^*)}{1-\delta(1-\frac{\Delta(x^*)}{2\sigma})} = \frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta)+\delta\Delta(x^*)}.
\end{aligned}
$$

$W(q,x')$ can then be written as:

$$
\begin{aligned}
W(q,x') &= (1-\delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q,x^*) \\
&\quad + \delta(1-\frac{\Delta(x^*)}{2\sigma})\frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta)+\delta\Delta(x^*)} - k|x^*-x'| \\
&= \max_x \mathbb{E}[y|x]\Big((1-\delta) + \delta(1-\frac{\Delta(x)}{2\sigma})\frac{2\sigma(1-\delta)}{2\sigma(1-\delta)+\delta\Delta(x)}\Big) \\
&\quad + \delta\Phi(q,x)\Big(\frac{\Delta(x)}{2\sigma} + (1-\frac{\Delta(x)}{2\sigma})\frac{\delta\Delta(x)}{2\sigma(1-\delta)+\delta\Delta(x)}\Big) - k|x-x'| \\
&= \max_x \mathbb{E}[y|x]\Big((1-\delta)\big(1+\frac{2\sigma\delta-\delta\Delta(x)}{2\sigma(1-\delta)+\delta\Delta(x)}\big)\Big) \\
&\quad + \delta\frac{\Delta(x)}{2\sigma}\Phi(q,x)\big(1+\frac{2\sigma\delta-\delta\Delta(x)}{2\sigma(1-\delta)+\delta\Delta(x)}\big) - k|x-x'| \\
&= \max_x \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q,x)}{2\sigma(1-\delta)+\delta\Delta(x)} - k|x-x'|.
\end{aligned}
$$

□

Lemma 4. *Suppose $x' \in [\underline{x},\overline{x}]$. Then $x_p(q,x') \in [\underline{x},\overline{x}]$.*

Proof. Note that our problem can be written as:

$$
\begin{aligned}
W(q,x') = \max_{x\in[0,1]} &\frac{2\sigma(1-\delta)}{2\sigma(1-\delta)+\delta\Delta(x)}\big(\mathbb{E}[y|x] - k|x-x'|\big) \\
&+ \frac{\delta\Delta(x)}{2\sigma(1-\delta)+\delta\Delta(x)}\big(\Phi(q,x) - k|x-x'|\big).
\end{aligned}
$$

First, since $\frac{\partial^2}{\partial q \partial x}\mathbb{E}[y|x] = f_g'(x) - f_b'(x) > 0$, since $\arg\max f_g(x) - k|x-x'| = \overline{x}$, and since $\arg\max f_b(x) - k|x-x'| = \underline{x}$, we have: $\arg\max_{\hat{x}\in[0,1]} \mathbb{E}[y|\hat{x}] - k|\hat{x}-x'| \in [\underline{x},\overline{x}]$. All $x < \underline{x}$ are dominated by $\underline{x}$ and all $x > \overline{x}$ are dominated by $\overline{x}$.

Next, I claim $\arg\max_{\tilde{x}\in[0,1]} \Phi(q,\tilde{x}) - k|\tilde{x} - x'| \in [\underline{x}, \overline{x}]$. Note that

$$\Phi(q,\tilde{x}) - k|\tilde{x} - x'| = qf_g(\overline{x}) + (1-q)f_b(\underline{x}) - k\big[q|\overline{x} - \tilde{x}| + (1-q)|\underline{x} - \tilde{x}| + |\tilde{x} - x'|\big].$$

This is concave and maximized at $\tilde{x} = x' \in [\underline{x}, \overline{x}]$. This means that any $\tilde{x} < \underline{x}$ is dominated by $\hat{x} \in [\underline{x}, x']$ and $\tilde{x} > \overline{x}$ is dominated by $\hat{x} \in [x', \overline{x}]$.

Finally, suppose by contradiction that $[\underline{x}, \overline{x}] \not\ni \hat{x}$, where $\hat{x}$ is the maximizer associated with $W(q, x')$. If $\hat{x} < \underline{x}$, we know that $\mathbb{E}[y|\hat{x}] - k|\hat{x} - x'| < \mathbb{E}[y|\underline{x}] - k|\underline{x} - x'|$ and $\Phi(q,\hat{x}) - k|\hat{x} - x'| < \Phi(q,\underline{x}) - k|\underline{x} - x'|$. Any convex combination of $\mathbb{E}[y|\underline{x}] - k|\underline{x} - x'|$ and $\Phi(q,\underline{x}) - k|\underline{x} - x'|$ is strictly larger than any convex combination of $\mathbb{E}[y|\hat{x}] - k|\hat{x} - x'|$ and $\Phi(q,\hat{x}) - k|\hat{x} - x'|$, meaning $\underline{x}$ dominates $\hat{x}$ in the original program. A symmetric argument shows that $\overline{x}$ would dominate $\hat{x} > \overline{x}$. $\square$

Lemma 5. *Suppose $k < f_g'(x_{LT}) < 2k$. There exists $c > 0$ such that: $f_g(\overline{x}) - k(\overline{x} - x_{LT}) \geq y_{LT} + c$, $f_b(\underline{x}) - k(x_{LT} - \underline{x}) + c > y_{LT}$ and $W(1/2, x_{LT}) < y_{LT} + c$. Hence, the set $LT = \{(q,x') : W(q,x') < y_{LT} + c\}$ has nonempty interior.*

Proof. We construct an upper bound for $W(1/2, x_{LT})$. First, I claim:

$$\frac{1}{2}\big(f_g(\overline{x}) - k(\overline{x} - x_{LT})\big) + \frac{1}{2}\big(f_b(\underline{x}) - k(x_{LT} - \underline{x})\big) - k(\overline{x} - x_{LT}) < y_{LT}$$
$$\frac{1}{2}\big(f_g(\overline{x}) - k(\overline{x} - x_{LT})\big) + \frac{1}{2}\big(f_b(\underline{x}) - k(x_{LT} - \underline{x})\big) - k(x_{LT} - \underline{x}) < y_{LT}.$$

Note that because $f_b(x)$ and $f_g(x)$ are symmetric about $x_{LT} = \frac{1}{2}$, $f_g(\overline{x}) - k(x_{LT} - \overline{x}) = f_b(\underline{x}) - k(x_{LT} - \overline{x})$. Hence, the first equation reduces to $f_g(\overline{x}) - 2k(\overline{x} - x_{LT}) < y_{LT}$, which follows since $f_g'(x_{LT}) < 2k$ and f_g is concave.

We now construct our upper bound for $W(1/2, x_{LT})$. $W(1/2, x_{LT}) =$

$$\begin{aligned}
&\max_{x\in[\underline{x},\overline{x}]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(1/2,x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x - x_{LT}| \\
&= \max_{x\in[\underline{x},\overline{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)}\big(\mathbb{E}[y|x] - k|x - x_{LT}|\big) + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)}\big(\Phi(1/2,x) - k|x - x'|\big) \\
&\leq \max_{x\in[\underline{x},\overline{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)}\big(\max_{\breve{x}} \mathbb{E}[y|\breve{x}] - k|\breve{x} - x_{LT}|\big) + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)}\big(\Phi(1/2,x) - k|x - x'|\big) \\
&= \max_{x\in[\underline{x},\overline{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)} y_{LT} + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)}\big(\Phi(1/2,x) - k|x - x_{LT}|\big) \equiv \max_{x\in[\underline{x},\overline{x}]} U(x).
\end{aligned}$$

$U(x)$ will be the objective associated with the upper bound. The third line holds since $\arg\max \mathbb{E}[y|\breve{x}]$ when $q = \frac{1}{2}$ is simply $x_{LT} = \frac{1}{2}$. Moreover, note that Φ does not depend on x at $q = 1/2$: $1/2(f_g(\overline{x}) - k(\overline{x} - x)) + 1/2(f_b(\underline{x}) - k(x - \underline{x})) = 1/2(f_g(\overline{x}) + f_b(\underline{x}) - k(\overline{x} - \underline{x}))$. Hence:

$$
\begin{aligned}
\Phi(1/2, x_{LT}) =& 1/2(f_g(\overline{x}) - k(\overline{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) > y_{LT} \\
&> 1/2(f_g(\overline{x}) - k(\overline{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\overline{x} - x_{LT}), \\
&\quad 1/2(f_g(\overline{x} - k(\overline{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(x_{LT} - \underline{x}).
\end{aligned}
$$

Let $\hat{x}_1$ and $\hat{x}_2$ be defined by the following:

$$
\begin{aligned}
y_{LT} &= 1/2(f_g(\overline{x}) - k(\overline{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\hat{x}_1 - x_{LT}) \qquad \hat{x}_1 \in (x_{LT}, \overline{x}) \\
y_{LT} &= 1/2(f_g(\overline{x}) - k(\overline{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(x_{LT} - \hat{x}_2) \qquad \hat{x}_2 \in (\underline{x}, x_{LT})
\end{aligned}
$$

which exist and are unique by continuity, strict monotonicity, and the Intermediate Value Theorem. Because we are taking convex combinations of y_{LT} and $\Phi - k|x - x_{LT}|$, where the weights depend on x as well, we will only want to place less weight on y_{LT} if $\Phi - k|x - x_{LT}| > y_{LT}$ (precisely on $[\hat{x}_2, x_{LT}]$ and $[x_{LT}, \hat{x}_1]$).

The derivative of $U(x)$ for $x > x_{LT}$ is:

$$
\frac{2\sigma(1-\delta)\delta\Delta'(x)(\Phi - k(x - x_{LT}) - y_{LT}) - 2\sigma(1-\delta)\delta\Delta(x)k - \delta^2\Delta(x)^2 k}{(2\sigma(1-\delta) + \delta\Delta(x))^2}.
$$

A nearly identical derivative for $x < x_{LT}$ is omitted. At $x = x_{LT}$, the right derivative is $\frac{\delta\Delta'(x)(\Phi - y_{LT})}{2\sigma(1-\delta)} > 0$. At $\hat{x}_1$, when $\Phi - k(\hat{x}_1 - x_{LT}) - y_{LT} = 0$, the derivative is $-\frac{\delta\Delta(x)k}{2\sigma(1-\delta)+\delta\Delta(x)} < 0$. The derivative is ≥ 0 when:

$$
\begin{aligned}
2\sigma(1-\delta)\delta\Delta'(x)(\Phi - k(x - x_{LT}) - y_{LT}) &\geq \delta\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))k \\
\Phi - k(x - x_{LT}) - y_{LT} &\geq \frac{\delta\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))}{2\sigma(1-\delta)\delta\Delta'(x)}k \\
\Phi - k(x - x_{LT}) &\geq y_{LT} + \frac{\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))}{2\sigma(1-\delta)\Delta'(x)}k \\
&= y_{LT} + k(x - x_{LT})\left(1 + \frac{\delta\Delta(x)}{2\sigma(1-\delta)}\right).
\end{aligned}
$$

The final line is strictly increasing in x with a zero at $x_{LT} < \hat{x}_1$. $\Phi - k(x - x_{LT})$ is strictly decreasing in x with a zero at $\hat{x}_1$. Hence there is a *unique* maximizer $\overline{x}^\star$ to this problem in $(x_{LT}, \hat{x}_1)$. Define c implicitly as $\Phi - k(\overline{x}^\star - x_{LT}) = y_{LT} + c$. [14] Then:

$$\begin{aligned} W(1/2, x_{LT}) \le \max_x U(x) &= y_{LT} + \frac{\delta\Delta(\overline{x}^\star)}{2\sigma(1-\delta) + \delta\Delta(\overline{x}^\star)}\left(k(\overline{x}^\star - x_{LT})\left(1 + \frac{\delta\Delta(\overline{x}^\star)}{2\sigma(1-\delta)}\right)\right) \\ &= y_{LT} + \frac{\delta\Delta(\overline{x}^\star)}{2\sigma(1-\delta) + \delta\Delta(\overline{x}^\star)} c < y_{LT} + c. \end{aligned}$$

Since $\overline{x}^\star > x_{LT}$, we also have $\Phi(1/2, x_{LT}) - k(\overline{x}^\star - x_{LT}) = y_{LT} + c \implies \Phi(1/2, x_{LT}) > y_{LT} + c$. Since $W(q, x')$ is continuous, the inequality $W(1/2, x_{LT}) < y_{LT} + c$ also holds in a *neighborhood* of $(1/2, x_{LT})$. This means that $LT : \{(q, x') : W(q, x') \le y_{LT} + c\}$ has nonempty interior. □

Proof of Solution to Leader's Repeated Problem

We proceed with the proof of the leader's problem. Lemmata 6 and 7 lay out stationarity features of the no-overthrow set $\mathrm{NR}(q, x')$. Lemma 8 illustrates what the leader would do if her choices were unconstrained. Theorem 1 generalizes Proposition 1.

First, in Lemma 6, we define $\tilde{W}(x_\ell, q, x')$ as the people's value from accepting the strategy $x_\ell(q, x')$ for all time. If the leader's strategy is only to implement x_{LT} for all time, the value of this objective is $y_{LT} - k|x_{LT} - x'|$. The following lemmata provide a closed form for this expression and show that, in equilibrium, this strategy has the property that $x_\ell(q, x') = x_\ell(q, x_\ell(q, x'))$. First, the people's utility from tolerating x_ℓ can be described recursively by:

$$\begin{aligned} \tilde{W}(x_\ell, q, x') = (1-\delta)\mathbb{E}[y|x_\ell(q, x')] - k|x_\ell(q, x') - x'| \\ + \delta\Bigg(\frac{\Delta(x_\ell(q, x'))}{2\sigma}\big(q\tilde{W}(x_\ell, 1, x_\ell(q, x')) \\ + (1-q)W(0, x_\ell(q, x'))\big) + \big(1 - \frac{\Delta(x_\ell(q, x'))}{2\sigma}\big)\tilde{W}(x_\ell, q, x_\ell(q, x'))\Bigg). \end{aligned}$$

Lemma 6. *Let $x_\ell(q, x')$ denote a policy for the leader such that $x_\ell(q, x_\ell(q, x')) = x_\ell(q, x')$. In this case,*

[14] Defining c using the derivative to the left of x_{LT} gives the same c, since Φ, $\Delta(x)$, and $k|x - x_{LT}|$ are symmetric about x_{LT}.

we have:

$$\tilde{W}(x_\ell, q, x') = \frac{2\sigma\big((1-\delta)\mathbb{E}[y|x_\ell(q,x')]\big) + \delta\Delta(x_\ell(q,x'))\tilde{\Phi}(q, x_\ell(q,x'))}{2\sigma(1-\delta) + \delta\Delta(x_\ell(q,x'))} - k|x_\ell(q,x') - x'|$$

and $\tilde{\Phi}(q,x) = q\tilde{W}(x_\ell, 1, x) + (1-q)\tilde{W}(x_\ell, 0, x)$.

Proof. Let $\tilde{W}(x_\ell, q, x')$ be the value for the people of accepting for all time a policy $x_\ell(q,x')$ satisfying the hypothesis. Utilizing a similar argument as Lemma 3 and the fact that $x_\ell(q,x') = x_\ell(q, x_\ell(q,x'))$ (meaning that, conditional on q remaining the same, adjustment costs are not incurred again), this can be rearranged as:

$$\tilde{W}(x_\ell, q, x') = \frac{2\sigma\big((1-\delta)\mathbb{E}[y|x_\ell(q,x')]\big) + \delta\Delta(x_\ell(q,x'))\tilde{\Phi}(q, x_\ell(q,x'))}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x_\ell(q,x') - x'|$$

where $\tilde{\Phi}$ is the expected utility from tolerating the leader conditional on the revelation of the true state:

$$\tilde{\Phi}(q,x) = q\tilde{W}(x_\ell, 1, x) + (1-q)\tilde{W}(x_\ell, 0, x).$$

As before, given a strategy with this property, the people have an incentive to overthrow if and only if $\tilde{W}(x_\ell, q, x') \geq W(q, x') - c$. □

Let $\mathrm{NR}(q,x')$ be the set of policies that weakly prevents overthrow (credibly):

$$\mathrm{NR}(q,x') = \big\{x \in [0,1] : \tilde{W}(x_\ell, q, x') \geq W(q,x') - c\big\}.$$

As before, the people overthrow the leader if and only if $x_\ell(q,x') \notin \mathrm{NR}(q,x')$.

Lemma 7. $x_\ell(q,x') = x_\ell\big(q, x_\ell(q,x')\big)$.

Proof. If $\mathrm{NR}(q,x')$ does not bind, the result follows immediately. Thus, suppose the constraint binds at each point in time. Consider first $q = 1$. Writing $x_0 = x'$, recursively define $x_t = x_\ell(1, x_{t-1})$. Note that $u_\ell(x)$ is increasing in x, meaning $x_t \geq x_{t-1}$ for all t. Suppose by contradiction that $x_t \geq x_{t-1}$ for all t and strictly for at least one t. Define $x^\dagger = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} x_t \in [0,1]$. Because

$f_g(x_t) - k(x_t - x_{t-1})$ is strictly concave and $u_\ell(x_t)$ is concave, we have that

$$f_g(x^\dagger) - k(x^\dagger - x_0) > \sum_{t=1}^{\infty}(1-\delta)\delta^{t-1}[f_g(x_t) - k(x_t - x_{t-1})]$$
$$u_\ell(x^\dagger) \geq \sum_{t=1}^{\infty}(1-\delta)\delta^{t-1}u_\ell(x_t),$$

i.e. $x^\dagger$ strictly relaxes the constraint for the people and weakly improves the leader's utility, contradicting that $x_\ell(q,x')$ maximized the leader's expected utility. A similar argument applies at $q=0$. This means that $x_\ell(q,x') = x_\ell\big(q, x_\ell(q,x')\big)$ at $q = 0,1$.

Next, we turn to q interior. Define $h(x) = (1-\delta)\mathbb{E}[y|x] + \frac{\delta\Delta(x)}{2\sigma}\tilde{\Phi}(q,x)$. Utilizing our previous notation with $x_t = x_\ell(q, x_{t-1})$ and $x_0 = x'$, suppose WLOG that $x_1 \neq x_2$. Recalling that $\tilde{W}(x_\ell, q, x_2)$ represents the continuation utility when the status quo is x_2, we have:

$$W(q,x_0) - c = h(x_1) - k|x_1 - x_0| + \delta\big(1 - \frac{\Delta(x_1)}{2\sigma}\big)\big(h(x_2) + \delta\big(1 - \frac{\Delta(x_2)}{2\sigma}\big)\tilde{W}(x_\ell, q, x_2) - k|x_2 - x_1|\big)$$
$$W(q,x_1) - c = h(x_2) + \delta\big(1 - \frac{\Delta(x_2)}{2\sigma}\big)\tilde{W}(x_\ell, q, x_2) - k|x_2 - x_1|.$$

Let $x^* = x_p(q, x_0)$. Since $\mathrm{NR}(q,x')$ is binding, we must have:

$$\underbrace{\frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x_0| - c}_{=W(q,x_0)-c} > h(x_2) + \delta\big(1 - \frac{\Delta(x_2)}{2\sigma}\big)\tilde{W}(x_\ell, q, x_2) - k|x_2 - x_0|$$

Since either $x_2 > x_1 > x^*$ or $x^* > x_1 > x_2$, adding $k|x_1 - x_0|$ to each side implies that:

$$\frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x_1| - c > \underbrace{h(x_2) + \delta\big(1 - \frac{\Delta(x_2)}{2\sigma}\big)\tilde{W}_{x_\ell}(x_2) - k|x_2 - x_1|}_{=W(q,x_1)-c}$$

meaning that $W(q,x_0) + k|x_1 - x_0| > W(q,x_1)$. But then:

$$\begin{aligned} W(q,x_1) &= \max_{x\in[0,1]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q,x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x - x_1| \\ &\geq \frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q,x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x_1| = W(q,x_0) + k|x_1 - x_0| > W(q,x_1), \end{aligned}$$

a contradiction. □

To formally characterize the leader's solution, we introduce the following notation:

$$\underline{x}_{LT}(q,x') = \max\{x \in \mathrm{NR}(q,x') : x \le x_{LT}\} \qquad \overline{x}_{LT}(q,x') = \min\{x \in \mathrm{NR}(q,x') : x \ge x_{LT}\}$$
$$\underline{x}(q,x') = \min\{x \in \mathrm{NR}(q,x')\} \qquad \overline{x}(q,x') = \max\{x \in \mathrm{NR}(q,x')\}.$$

$\underline{x}_{LT}$ and $\overline{x}_{LT}$ correspond to the *closest* policies to the learning trap preventing overthrow. $\overline{x}$ and $\underline{x}$ are the lowest and highest policies preventing overthrow altogether. As before, we also have:

$$\underline{u}(x') \equiv u_\ell(x_\ell(0,x')) < u_\ell(y_{LT}) < u_\ell(x_\ell(1,x')) \equiv \overline{u}(x').$$

The Bellman describing the leader's optimal solution $x_\ell(q,x')$ is then:

$$V(q,x') = \max_{x \in \mathrm{NR}(q,x')} (1-\delta^\ell)u_\ell(x) + \delta^\ell\left(\frac{\Delta(x)}{2\sigma}\Psi(q,x) + (1-\frac{\Delta(x)}{2\sigma})V(q,x)\right)$$

where $\Psi(q,x) = q\overline{u}(x) + (1-q)\underline{u}(x)$.

Lemma 8. *Suppose that the leader can implement any $x \in [x_\ell(0,x'), x_\ell(1,x')]$. Then, there exist thresholds $\underline{q} < \overline{q}$ such that: for $q \le \underline{q}$, the derivative of the objective in the maximization problem is positive for $x < x_{LT}$ and negative for $x > x_{LT}$; for $q \in [\underline{q},\overline{q}]$, the derivative of the objective is positive for $x < x_{LT}$ and positive for $x > x_{LT}$; and for $q \ge \overline{q}$, the derivative of the objective is negative for $x < x_{LT}$ and positive for $x > x_{LT}$.*

Proof. Let $X_{unc} = [x_\ell(0,x'), x_\ell(1,x')]$. The leader's Bellman in the unconstrained case can be written as:

$$\tilde{V}(q) = \max_{x \in X_{unc}} (1-\delta^\ell)u_\ell(x) + \delta^\ell\frac{\Delta(x)}{2\sigma}(q\overline{u}(x) + (1-q)\underline{u}(x)) + \delta^\ell\left(1-\frac{\Delta(x)}{2\sigma}\right)\tilde{V}(q).$$

Let $\Psi(q,x) = q\overline{u}(x) + (1-q)\underline{u}(x)$. Replicating an earlier argument implies

$$\tilde{V}(q) = \max_{x \in X_{unc}} \frac{2\sigma(1-\delta^\ell)u_\ell(x) + \delta^\ell\Delta(x)\Psi(q,x)}{2\sigma(1-\delta^\ell) + \delta^\ell\Delta(x)}.$$

The numerator of the derivative with respect to x for $x \neq x_{LT}$ is:

$$= 4\sigma^2(1-\delta^\ell)^2 u_\ell'(x) + 2\sigma\delta^\ell(1-\delta^\ell)(\Delta(x)u_\ell'(x) + \Delta'(x)\Psi(q,x) - \Delta'(x)u_\ell(x)).$$

Observe that $\Delta(x)$ is the absolute value of a linear function with zero at $x_{LT} = \frac{1}{2}$, which we write as $\Delta(x) \equiv \xi|x - x_{LT}|$.

This means that for $x > x_{LT}$, the derivative of the objective is negative if and only if

$$2\sigma(1-\delta^\ell)u_\ell'(x) \leq \delta^\ell\big(\Delta'(x)u_\ell(x) - \Delta(x)u_\ell'(x) - \Delta'(x)\Psi(q,x)\big)$$
$$2\sigma(1-\delta^\ell)u_\ell'(x) \leq \delta^\ell\xi\big(u_\ell(x) - u_\ell'(x)(x - x_{LT}) - \Psi(q,x)\big)$$

By concavity, $u_\ell(x) - u_\ell'(x)(x - x_{LT})$ is equal to x_{LT} at $x = x_{LT}$ and is weakly increasing for $x > x_{LT}$.[15] By weak concavity, the LHS below is decreasing and the RHS increasing:

$$2\sigma(1-\delta^\ell)u_\ell'(x) \leq \delta^\ell\xi\big(u_\ell(x) - u_\ell'(x)(x - x_{LT}) - \Psi(q,x)\big) \tag{12}$$

If $\sigma(1-\delta^\ell)$ is small, since $\underline{u}(x') < x_{LT} \leq u_\ell(x) - u_\ell'(x)(x - x_{LT}) < \overline{u}(x')$, there is a threshold $\underline{q}$ such that the inequality satisfies strictly if and only if $q < \underline{q}$. Moreover, for any $\sigma(1-\delta^\ell)$, as $\underline{u}(x)$ grows more negative, $-\Psi(q,x)$ grows large, which can also allow the inequality to be satisfied. A similar argument shows that for $x < x_{LT}$, the derivative is positive when:

$$2\sigma(1-\delta^\ell)u_\ell'(x) \geq \delta^\ell\xi\big(\Psi(q,x) - \big(u_\ell(x) - u_\ell'(x)(x - x_{LT})\big)\big) \tag{13}$$

When (12) holds with equality, its RHS is strictly positive, meaning the RHS of (13) is strictly negative. Letting $\overline{q}$ be the point where (13) holds with equality, we thus have that $\underline{q} < \overline{q}$. Hence: for $x > x_{LT}$, the derivative of the objective is strictly negative when $q < \underline{q}$ and strictly positive when $q > \underline{q}$, with equality at $q = \underline{q}$; for $x < x_{LT}$, the derivative of the objective is strictly positive when $q < \overline{q}$ and strictly negative when $q > \overline{q}$, with equality at $q = \overline{q}$. This completes the lemma. □

Theorem 1. *Suppose $\sigma(1-\delta^\ell)$ is small or $\underline{u}(x')$ is sufficiently low. There exist thresholds $\underline{q} < \overline{q} \in (0,1)$*

[15] To see this, note that the derivative of this expression with respect to x is $u_\ell'(x) - u_\ell''(x)(x - x_{LT}) - u_\ell'(x) = -u_\ell''(x)(x - x_{LT}) \geq 0$.

such that:

1. *If $q \leq \underline{q}$, the leader plays the* closest policy *to the learning trap (which may be* greater *or* less *than x_{LT}).*

2. *If $q \in [\underline{q}, \overline{q}]$, the leader plays either the highest policy below the learning trap or her highest feasible policy.*

3. *If $q > \overline{q}$, the leader either plays her highest feasible policy or lowest feasible policy.*

The hypotheses of the Theorem are described in terms of the above notation as follows.

- If $q \leq \underline{q}$, then $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\overline{x}_{LT}(q, x')$.

- If $q \in [\underline{q}, \overline{q}]$, then $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\overline{x}(q, x')$.

- If $q > \overline{q}$, = then $x_\ell(q, x') = \overline{x}(q, x')$ or $\underline{x}(q, x')$.

Proof. We use the lemmata above to prove the original theorem; the solution can be written:

$$V(q, x') = \max_{x \in \mathrm{NR}(q, x')} \frac{2\sigma(1 - \delta^\ell) u_\ell(x) + \delta^\ell \Delta(x) \Psi(q, x)}{2\sigma(1 - \delta^\ell) + \delta^\ell \Delta(x)}$$

where, replicating the previous lemma's argument, we have $\Psi(q, x) = q\overline{u}(x) + (1 - q)\underline{u}(x)$. If $q \leq \underline{q}$, the derivative for $x > x_{LT}$ is negative and the derivative for $x < x_{LT}$ is positive, meaning the leader is attracted to the learning trap. Hence, $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\overline{x}_{LT}(q, x')$. If $q \in [\underline{q}, \overline{q}]$, the solution is either $\underline{x}_{LT}(q, x')$ or some $x \geq x_{LT}$. If $q \geq \overline{q}$. the derivative for $x > x_{LT}$ is positive and the derivative for $x < x_{LT}$ is negative. Hence the optimal solution is either $\overline{x}(q, x')$ or $\underline{x}(q, x')$. □

Additional Results

Model with Non-Uniform Distribution of ϵ_t

Proposition 4. *Suppose the graph of* $\mathrm{NR}(q)$ *is convex. There exists a threshold* $\underline{q} \in (0,1]$ *such that:*

1. *if* $q \leq \underline{q}$, $x_\ell(q) = \overline{x}(q)$. *The leader plays the highest policy preventing overthrow.*
2. *If* $q > \underline{q}$, *the leader implements* $x_\ell(q) \in [\min\{x_{LT}, \underline{x}(q)\}, \overline{x}(q)]$. *Moreover, for* δ *high,* $x_\ell(q) < \overline{x}(q)$. *I.e., she implements a policy closer to the learning trap than her maximal policy* $\overline{x}(q)$.

Proof. Suppose q is such that $\overline{x}(q) \leq 1$. Because the graph of $\mathrm{NR}(q) = [q - \sqrt{c}, \min\{q + \sqrt{c}, 1\}]$ is convex, by Theorem 9.8 in Stokey et al. (1989), $V(q)$ is concave in q.

Let $\underline{q} \in (0,1)$ solve $\underline{q} + \sqrt{c} = x_{LT}$. Suppose $q \leq \underline{q}$. Note that, on this range, an increase from x to x' has two effects. First, it increases flow utility: $u_\ell(x') > u_\ell(x)$. Second, because $q \leq \underline{q}$, we must have $x_{LT} \geq x' > x$, so $|x - x_{LT}| > |x' - x_{LT}|$, meaning $\tau(q'|q, x')$ is a mean-preserving contraction of $\tau(q'|q, x)$. This means that, because $V(q)$ is concave in q, $\int V(q')\tau(q'|q,x') > \int V(q')\tau(q'|q,x)$. Hence, the leader will pursue $\overline{x}(q)$, which is both the highest possible policy and the closest policy to the learning trap.

Next, suppose $q > \underline{q}$. By the previous argument, all feasible $x < x_{LT}$ are dominated by x_{LT}. Now, above x_{LT}, an increase from x to x' has two effects. On the one hand, flow utility increases. On the other, because V is concave, it *decreases* the value of continuation utility, i.e. $\int V(q')\tau(q'|q,x') < \int V(q')\tau(q'|q,x)$. For δ large, this second informational effect will tend to outweigh the former flow effect, suggesting that $\min\{\underline{x}(q), x_{LT}\} \leq x_\ell(q) < \overline{x}(q)$. □

Three States of the World

This section establishes an analogue of Lemma 8 for three states of the world, since deriving properties of $\mathrm{NR}(q)$ is challenging. The three states of the world are g, m, b. The data-generating process in state i is given by f_i:

$$\begin{aligned} f_g(x) \qquad & y_t = -(x_t - 1)^2 + \epsilon_t \\ f_m(x) \qquad & y_t = -(x_t - \tilde{x}_m)^2 + \epsilon_t \\ f_b(x) \qquad & y_t = -x_t^2 + \epsilon_t. \end{aligned}$$

for $\tilde{x}_m \in (0,1)$ and $\epsilon_t \sim \mathcal{U}[-\sigma, \sigma]$. We assume $f_i(x) - f_j(x)$ each single-cross at some x_{LT}^{ij} for all $i \neq j$. The state variables are q_g (belief that g holds) and q_m (belief that m holds). We fix $q_g, q_m > 0$ and $q_g + q_m < 1$ for the remainder of the analysis. Beginning at a vector of states (q_g, q_m), we can write the following expected values of the problem, as well as the probabilities with which they occur (which are all functions of x).

- V^{gmb}: initial value of the problem at (q_g, q_m).
- V^g: value of problem if g is true; $(q_g, q_m) \to (1, 0)$; occurs with prob. p_g. Solution is x_ℓ^g.
- V^m: value of problem if m is true; $(q_g, q_m) \to (0, 1)$; occurs with prob. p_m Solution is x_ℓ^m.
- V^b value of problem if b is true; $(q_g, q_m) \to (0, 0)$; occurs with prob. p_b. Solution is x_ℓ^b.
- V^{gb} value of problem if only m is not true; $(q_g, q_m) \to (\frac{q_g}{1-q_m}, 0)$; occurs with prob. p_{gb}.
- V^{gm} value of problem if only b is not true; $(q_g, q_m) \to (\frac{q_g}{q_g+q_m}, \frac{q_m}{q_g+q_m})$; occurs with prob. p_{gm}.
- V^{mb} value of problem if only g is not true; $(q_g, q_m) \to (0, \frac{q_m}{1-q_g})$; occurs with prob. p_{mb}.

Each of these probabilities can be computed explicitly in terms of the beliefs and x; for example, $p_g(x)$ is given by $q_g \cdot \frac{|f_g(x) - f_b(x)|}{2\sigma} \cdot \frac{|f_g(x) - f_m(x)|}{2\sigma}$.

Note that $x_\ell^b < x_{LT}^{gb} < x_\ell^g$ and $x_\ell^b < x_{LT}^{mb} < x_\ell^m$. Fixing $1 - q_m - q_b$, suppose the difference between $V^b = u_\ell(x_\ell^b)$ and $V^m = u_\ell(x_\ell^m)$ is sufficiently large. Applying Lemma 8 shows:

- $V^{gb} = u_\ell(x_{LT}^{gb})$: suppressing the risk of information revelation that could reveal b as true, the leader will choose a "pairwise" learning trap x_{LT}^{gb}
- $V^{mb} = u_\ell(x_{LT}^{mb})$: suppressing the risk of information revelation that could reveal b as true, the leader will choose a "pairwise" learning trap x_{LT}^{mb}

This gives us the following result:

Lemma 9. *Suppose that $q_g, q_m > 0$ and $q_g + q_m < 1$; the leader's policies are constrained if g, m, or b are revealed as true, but can otherwise play any policy in $[x_\ell^b, x_\ell^g]$; and V^b is sufficiently low. Then, the leader either implements a learning trap policy x_{LT}^{gb} or x_{LT}^{mb}.*

Proof. Replicating earlier arguments, the leader's problem can be written as a convex combination of flow and continuation values:

$$V^{gmb} = \max_{x \in [x_\ell^b, x_\ell^g]} \frac{(1-\delta)u_\ell(x) + \delta\big(p_g V^g + p_m V^m + p_b V^b + p_{gb} V^{gb} + p_{gm} V^{gm} + p_{mb} V^{mb}\big)}{(1-\delta) + \delta\big(p_g + p_m + p_b + p_{gb} + p_{gm} + p_{mb}\big)}$$
$$V^{gmb} = \max_{x \in [x_\ell^b, x_\ell^g]} \frac{(1-\delta)u_\ell(x) + \delta\big(p_g V^g + p_m V^m + p_b V^b + p_{gb} u_\ell(x_{LT}^{gb}) + p_{gm} V^{gm} + p_{mb} u_\ell(x_{LT}^{mb})\big)}{(1-\delta) + \delta\big(p_g + p_m + p_b + p_{gb} + p_{gm} + p_{mb}\big)}.$$

Since we assume V^b is sufficiently low, the leader will do anything in her power to suppress revealing that b is true, i.e. will try her best to set $p_b = 0$. p_b is expressed as:

$$p_b = (1 - q_g - q_m)\frac{|f_g(x) - f_b(x)|}{2\sigma} \cdot \frac{|f_m(x) - f_b(x)|}{2\sigma}.$$

With probability $1 - q_g - q_m$, b is true. Then, with probability $\frac{|f_g(x)-f_b(x)|}{2\sigma}$, state b is separable from state g and with probability $\frac{|f_g(x)-f_b(x)|}{2\sigma}$ it is separable from state m. Because the left and right derivatives of p_b are always nonzero, the only way to set p_b equal to zero is either to set $f_g(x) = f_b(x)$ by playing x_{LT}^{gb}, or to set $f_m(x) = f_b(x)$ by implementing x_{LT}^{mb}. In the former case, either m is revealed as the state or the people are uncertain about the state being b or g; the value for the leader is:

$$\frac{(1-\delta)u_\ell(x_{LT}^{gb}) + \delta\big(p_m V^m + p_{gb} u_\ell(x_{LT}^{gb})\big)}{(1-\delta) + \delta\big(p_m + p_{gb}\big)}.$$

In the latter case, either g is revealed as the state or the people are uncertain between b and m; the value for the leader is:

$$\frac{(1-\delta)u_\ell(x_{LT}^{mb}) + \delta\big(p_g V^g + p_{mb} u_\ell(x_{LT}^{mb})\big)}{(1-\delta) + \delta\big(p_g + p_{mb}\big)}.$$

If q_g is sufficiently large, the leader will choose to implement x_{LT}^{mb}, which favors her in expectation;

otherwise, she will choose x_{LT}^{gb}. $\square$

Finally, the terminal history of the model always reduces to one or two states of the world. Suppose the leader implements x_{LT}^{gb}, allowing information revelation only about whether $\{g, b\}$ or $\{m\}$ is true. In the former case, we arrive at the baseline two-state model. On the equilibrium path, uncertainty is resolved only upon the revelation of states to which the leader is not averse.

Model with Ambiguity

Proposition 5. *Let $\mathcal{Q} \subseteq [0,1]$ be a compact set of constituent beliefs such that $\mathcal{Q} \cap (0,1) \neq \varnothing$. Suppose the leader operates as a maxmin expected utility maximizer and δ is large. Let $x_\ell(\mathcal{Q})$ be the leader's optimal policy conditional on believing the people's belief q lies in $\mathcal{Q}$. Then:*

- *if $x_{LT} \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = x_{LT}$;*
- *if $x_{LT} \notin \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$ but there exists x such that $x \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q})$ is the highest policy that prevents overthrow;*
- *if there does not exist $x \in [0,1]$ such that $x \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = 1$.*

Proof. Define $\mathrm{NR}(\mathcal{Q}) \equiv \bigcap_{q \in \mathcal{Q}} \mathrm{NR}(q)$. Because $\mathrm{NR}(q)$ is closed for each q, $\mathrm{NR}(\mathcal{Q})$ is also closed, but may be empty. As before, we normalize all flow utilities to $(1-\delta)u_\ell(x)$.

Suppose the leader implements $x \neq x_{LT}$. Since the anti-leader state of the world realizes with positive probability, the leader evaluates her continuation value as $\underline{u}$, meaning her value from pursuing x is $(1-\delta)u_\ell(x) + \delta\underline{u}$. This is maximized at $u_\ell(1) = 1$. If the leader implements $x = x_{LT}$, no information is revealed. Thus, if $x_{LT} \in \mathrm{NR}(\mathcal{Q})$ today, it is also in no-overthrow constraint tomorrow, meaning the utility from pursuing x_{LT} is x_{LT}. Since δ is large, $x_{LT} \geq (1-\delta) + \delta\underline{u}$. Hence, if $x_{LT} \in \mathrm{NR}(\mathcal{Q})$, the leader pursues x_{LT}.

If $x_{LT} \notin \mathrm{NR}(\mathcal{Q})$ but there exists x such that x such that $x \in \mathrm{NR}(q)$ for all $q \in \mathcal{Q}$, the leader's value is given by $\max_{x \in \mathrm{NR}(\mathcal{Q})}(1-\delta)u_\ell(x) + \delta\underline{u}$. This is maximized at $x = \max\{\mathrm{NR}(\mathcal{Q})\}$.

Finally, suppose $\mathrm{NR}(\mathcal{Q})$ is empty. This means any x the leader pursues could risk overthrow, meaning her policy choices do not matter, and pursuing $x = 1$ is an equilibrium. $\square$

External Information Revelation

Proposition 6. *$\underline{q}$ is decreasing in η. The leader plays the learning trap for fewer values of q.*

Proof. The objective of an unconstrained problem here can be written as:

$$\max_{x\in[0,1]} \frac{2\sigma(1-\delta)u_\ell(x) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)\Psi(q)}{2\sigma(1-\delta) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)}.$$

The numerator of the derivative can be expressed as:

$$\begin{aligned}
&\big(2\sigma(1-\delta) + \delta(1-\eta)\Delta(x) + 2\sigma\delta\eta\big)\big(2\sigma(1-\delta)u_\ell'(x) + \delta(1-\eta)\Delta'(x)\Psi(q)\big)\\
&-\delta(1-\eta)\Delta'(x)\big(2\sigma(1-\delta)u_\ell(x) + \delta\big((1-\eta)\Delta(x) + 2\sigma\eta\big)\Psi(q)\big)\\
&= 4\sigma^2(1-\delta)^2u_\ell'(x) + 2\sigma\delta(1-\delta)(1-\eta)\big(\Delta(x)u_\ell'(x) + \Delta'(x)\Psi(q) - \Delta'(x)u_\ell(x))\big)\\
&+ 2\sigma\delta\eta\big(2\sigma(1-\delta)u_\ell'(x) + \delta(1-\eta)\Delta'(x)\Psi(q) - (1-\eta)\Delta'(x)\Psi(q)\big)\\
&= 4\sigma^2(1-\delta)^2u_\ell'(x) + 2\sigma\delta(1-\delta)(1-\eta)\big(\Delta(x)u_\ell'(x) + \Delta'(x)\Psi(q) - \Delta'(x)u_\ell(x))\big)\\
&+ 4\sigma^2\delta(1-\delta)\eta u_\ell'(x).
\end{aligned}$$

For $x > x_{LT}$, the derivative is negative, replicating a previous lemma, if:

$$2\sigma(1-\delta)u_\ell'(x) + 2\sigma\delta\eta u_\ell'(x) \le \delta(1-\eta)\xi\big(u_\ell(x) - u_\ell'(x)(x - x_{LT}) - \Psi(q)\big)$$

where $\underline{q}$ is the unique point that solves this equation with equality. For each x, the LHS is increasing in η, and the RHS is decreasing in η. This means that as η increases from 0, $\underline{q}$ shifts to the left. The equation defining $\overline{q}$ can be written as:

$$2\sigma(1-\delta)u_\ell'(x) + 2\sigma\delta\eta u_\ell'(x) = \delta(1-\eta)\xi\big(\Psi(\overline{q}) - \big(u_\ell(x) - u_\ell'(x)(x - x_{LT})\big)\big)$$

An increase in η likewise causes the LHS to increase and the RHS to decrease, causing $\overline{q}$ to increase.

□

Additional Case Study Material

Extreme Equilibrium of Model: Stolypin Reforms

The main text argues that when there was uncertainty about whether liberal reform was aligned with constituents' interests, the Russian Tsar pursued a learning trap; indeed, few radical modifications to the post-1861 arrangement occurred over the ensuing decades. This section argues that when there was more *certainty* that liberal reform would benefit Russians, the Tsar's government experimented with private property under the programs of Prime Minister Pyotr Stolypin beginning in 1906.

These reforms were motivated by two forces that represent a large shock to (subjective) beliefs in favor of liberal reform and the pro-leader state. The first was the culmination of investigations into the affairs of the peasantry led by Sergei Witte, the Russian Minister of Finance, whose findings — along with independent recommendations of the Ministry of Internal Affairs — favored the privatization of property and "individualized farming" (Yaney, 1964, 278). The second was the Russian Revolution of 1905 — peasant uprisings that led to the establishment of (limited) constitutional monarchy (Polunov et al., 2015, 223) and the suspension of households' redemption payments, making the reformation of social order — changing the relationship of the peasantry to the land — necessary to complement these new political institutions. Indeed, the reforms existed "within the context of other legislative and organizational measures that Nicholas II's government introduced" (Pallot, 1999, 3). The confluence of these forces meant that "a sufficient number of officials in Nicholas II's government shared this conviction [that individualized farming should be introduced into the countryside] for it to find its way into official state policy" (Pallot, 1999, 4).

Stolypin's 1906 Reform gave households the choice between holding property in private or communal tenure. Households could leave the commune while maintaining allotments or an equivalent parcel of land, overseen by a state-mediated bargaining process. The dissolution of the commune's power was accompanied by a reduction in the enforcement of mobility restrictions. Stolypin indeed believed that peasants who found themselves landless would find industrial work in the cities. From 1906 until the outbreak of the First World War, over one third of peasant households withdrew from the commune (Polunov et al., 2015, 231-32).

Stolypin's reforms were seen as experiments. The means of achieving land enclosure and alignment of rural social order with politically liberal institutions evolved as officials learned about peasants' willingness to take up the reform and grappled with administration (Yaney, 1964, 275-276, 283). The culmination of the process would, ideally, result in economic growth for Russia, and "[o]nce enlightened through education and example to the possibilities of organizing their farms differently, it was thought that the majority of Russia's peasants would greet the reform enthusiastically" (Pallot, 1999, 5), creating "an independent, prosperous husbandman, a stable citizen of the land" (1).

Thus, through the lens of the model, we can view the Stolypin reforms as departures from the learning trap triggered by an exogenous change in constituents' beliefs that favored the applicability of liberal reform to Russia. While the eventual course of the reform was cut short by the First World War, many argued that these reforms put Russia on the path towards economic development in imitation of the Western European model. Dower and Markevich (2019) argue that, indeed, the reform had a positive effect on the productivity of land-use in its short lifespan.

Alternative Explanations for Learning Trap

The literature discusses two main *non-informational* arguments motivating the "middle ground" achieved by the Tsar's 1861 reforms. First, this mixture may have balanced the desires of elites with those of the state. Second, Russia's lack of state capacity may have prevented more radical reform. In this section, I show that while these factors undoubtedly affected the emancipation reforms, the threat of informational revelation still shaped the Tsar's policy choices, and the explanation of a learning trap can be seen as complementary to these other stories.

Elite Power The landed nobility's commitment to Serfdom constrained previous attempts to abolish Serfdom. Could the post-1861 reforms have simply been an outcome of bargaining between a liberal State and pro-Serfdom nobles?

On the one hand, the commune may have been logistically preferable for nobles to deal with civil and property disputes (Khristoforov and Gilley, 2016, 12). Labor allocation and rental payments may have been easier for landlords to organize when serfs were held collectively responsible (Dennison, 2014, 257). Landlords often disapproved of arrangements forcing them to give land to peasants, and extensively bargained over the size of government compensation during the years of the reform (Moon, 2014, chaps. 7-8). Nevertheless, the constraint binding the government *to the commune* was far weaker in the 1850s than prior.

More broadly, nobles in the 1850s were ideologically diverse, and many of their preferences aligned with the liberal aspirations of the Tsar. Most educated Russians believed that "the sanctity of private property was the basis of the political and social order" (Field, 1976, 58). Liberals often supported abolishing Serfdom and endowing peasants with land, but even some conservative nobles still wanted peasants to rent and work noble land in a competitive market. Many critics of the post-1861 system — at times more critical of Serfdom than the government's own reformers — contrasted Russia's system of land tenure with the more desirable British case, "where large-scale land tenure and local self-government controlled by the aristocracy allegedly guaranteed economic prosperity and political stability" (Khristoforov, 2009, 58, 63-65). Khristoforov (2022) goes as far as to argue that some form of "private property individualism" was in the interest of both landlords and the government.[16]

Finally, a sufficient bloc of the nobility may not have wanted to challenge the Tsar; they "did not want political change. . . They wanted the tsar's favor. . . At each stage in the development of the government's program, the nobility swallowed its objections or stated them obliquely" (Field, 1976, 362). Thus, not only may liberal reform not have been "at a cost" to the nobility, even if it was, it is not clear that they had much power to usurp the Tsar's decision-making.

State Capacity State capacity undoubtedly constrained the Tsar's reform ambitions. Emancipating and endowing peasants with property, compensating nobles, setting up a cadastre, and securing property rights would have been expensive. Depressed stock prices and banking crises in the 1850s drained coffers (Dennison, 2020, 197), meaning the government had to look to the gentry for help. Landlords then played a large role in commune politics after emancipation (Dennison, 2020); outsourcing dispute resolution and tax collection to the commune was logistically easier than protecting individual household rights (Khristoforov (2022, S163), Dennison (2011)).

Puzzles still arise from ascribing the intermediacy of the reform solely to weak administration. First, before and after the 1850s, the government could have relaxed the commune's power on State lands, where elites had a much weaker hold. Instead, the institutional experience of

[16] One reason was that a liberal system could oppose socialism.

the State peasantry faced few changes before and after 1861, and outcomes of ex-Serf and State peasants converged by 1900 (Nafziger, 2012). Second, the State made few attempts to demarcate property. Officials wanted to tie peasants to the land, knew private property was easier to abandon through sale or mortgage, and in fact refused to separate the tax liabilities of some peasants who paid off their redemption obligations early (Khristoforov, 2007, 30). Dennison (2006) shows how the post-1861 commune sustained a pre-Serfdom informal economy of social networks and patronage, even though the state could have slowly introduced "universal enforcement of contracts and property rights" (89).

Third, the state's nebulous attitude towards demarcation emanated in part from an informational desire to limit market activity. The results of a project attempting to construct a land cadastre prior to 1861 under Kiselev were never clearly revealed or publicly accessible, likely because the results may not have favored rationalization of property (Khristoforov and Gilley, 2016, 9, 11). This point is closely related to the model's extension with *private information* where, if a leader had a choice, she would choose *not* to set up institutions that collected information.

Fourth, while the Tsar's committees "recognized [the commune] as a temporary fact" (Khristoforov, 2007, 30), Alexander II never took further steps towards dissolution of the commune, and his son infamously strengthened its enforcement powers [17] (Wortman, 1989, 21). Finally, the reforms were intended to *increase* fiscal revenue and expand state capacity. If liberal reform were truly beneficial for peasant income, maintaining the commune seems surprising. However, the government's belief that liberal reform could have shrunk agricultural output and caused land alienation suggests that uncertainty played a crucial role in the post-1861 mixture.

[17] For example, "[a]dditional legislation in 1893 explicitly forbade all sales of allotment land to non-peasants" and required a two-thirds majority of communal assembly votes to redeem or release landholdings (Nafziger, 2010, 383).